

title: Logbook BL4S 2026, August Part I
tags: BL4S, Logbook, 2026, August Part I

Logbook BL4S 2026, August Part I

Run Planning

Calibration runs of the calorimeters

-1 GeV, -2 GeV, -3 GeV, Trigger S0&S1

this is is not complete, and can be wrong! we love your enthusiasm but please don't make edits on the logbook without supervision of your SS.

- ✓ CAL0; 1786782465 , 1786787514 , 1786788903
- ✓ CAL1; 1786800039, 1786799056, 1786798201
- ✓ CAL2; 1786781408, 1786780617, 1786779710
- ✓ CAL4; 1786657998, 1786662364, 1786660965
- ✓ CAL5; 1786794828, 1786796090, 1786797061
- ✓ CAL7; 1786836376 , 1786837396, 1786838309
- ✓ CAL8; 1786714515, 1786713224, 1786711761
- ✓ CAL9; 1786741940, 1786739830, 1786726168
- ✓ CAL10; 1786793439, 1786791937, 1786790499
- ✓ CAL11; 1786747305, 1786746501, 1786745756
- ✓ CAL12; 1786801341, 1786802590 , 1786803907
- ✓ CAL13; 1786743090, 1786744038, 1786744833
- ✓ CAL14; 1786748346, 1786750147, 1786750887
- ✓ CAL17; 1786807867, 1786806354, 1786805198
- ✓ CAL18; 1786864539, 1786840283, 1786839384
- ✓ CAL19; 1786723178, 1786724681, 1786725412

Pedestal runs of the calorimeters

Fake signal as trigger

cal0 should be taken as ch16, not as ch0.

- ☑ HV ON, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : 1786828994
- ☑ HV OFF, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : 1786829188

Calibration of the clustering algorithm

Trigger S0&S1&FS0&FS1

For central two calorimeters, move the DESY table 1 2 3 4 5 cm at 1 2 3 GeVs and observe the clustering code.

- ☑ CAL4
- ☐ CAL18

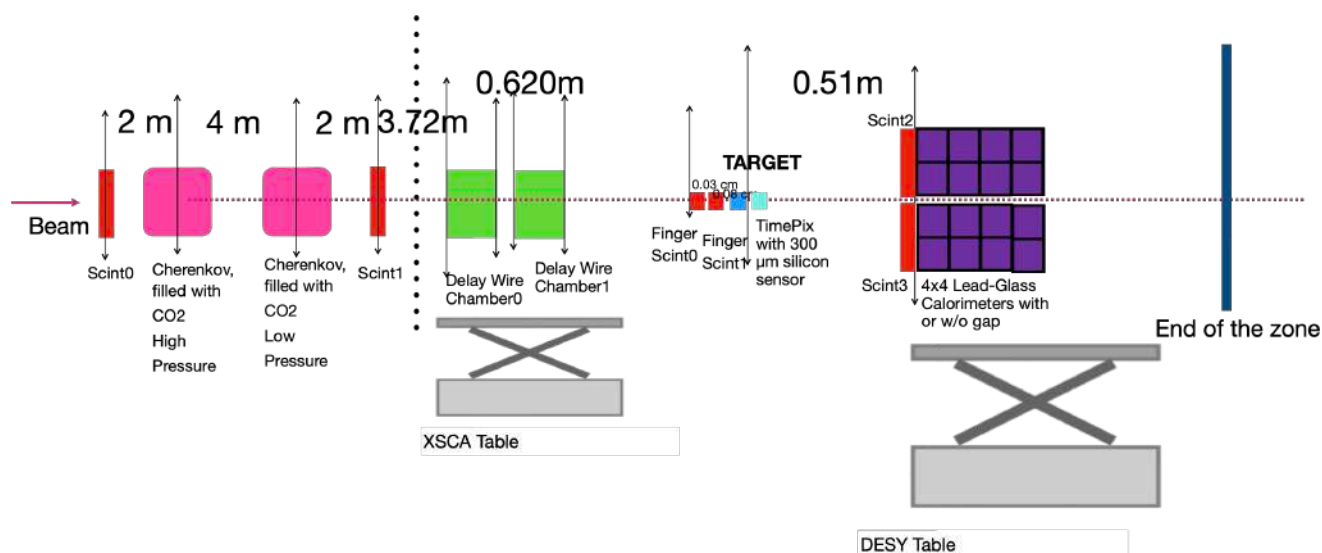
Physics Runs

- ☐ Trigger S0&S1&FS0&FS1. At 1.5, 2.0, 2.5, 3 GeV, with graphite of 1 and 3 cm thickness we should get data with lead glass calorimeters at 50 cm TBC for other energies than the 3 GeV.

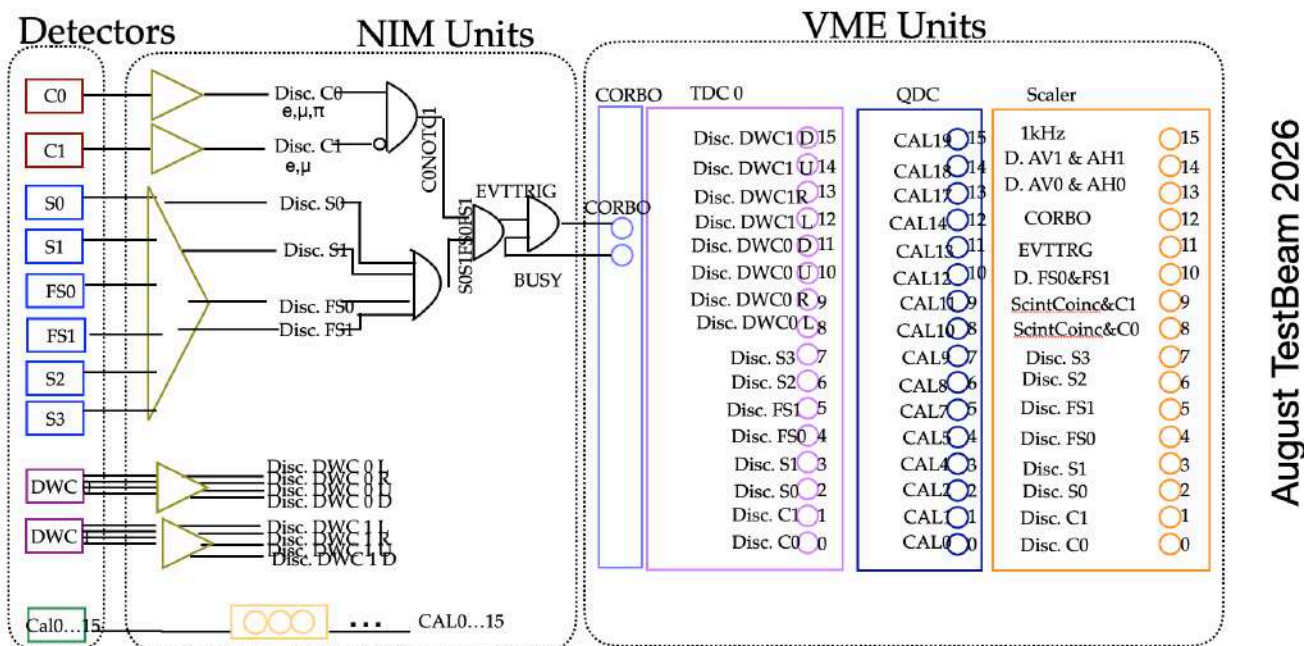
...

Experimental Layout

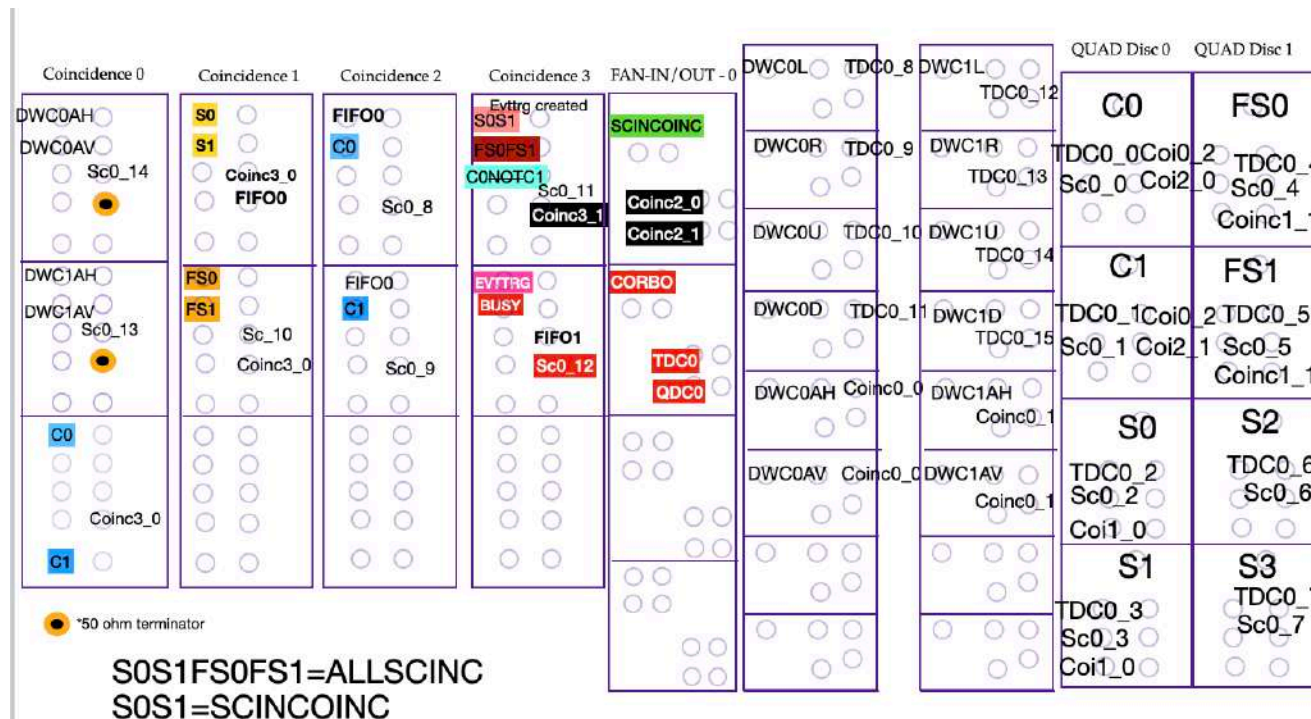
Distances are to be measured



VME Scheme



NIM Scheme



CAL CONFIGURATIONS:

CAL10 | CAL5 | CAL1 | CAL12

CAL17 | CAL4 | CAL18 | CAL7

CAL0 | CAL2 | CAL14 | CAL11

CAL8 | CAL19 | CAL9 | CAL13

DESY Positioning For Calibration Measurements:

- CAL 4: H:31.472 V: -3.696
 - CAL 17: H: 21.699 V: -3.696
 - CAL 18: H: 41.754 V: -3.384
 - CAL 7: H: 51.774 V: -3.439
 - CAL 12: H: 52.219 V: -13.080
 - CAL 1: H: 42.114 V: -12.986
 - CAL 5: H: 31.854 V: -12.986
 - CAL 10: H: 21.688 V: -13.168
 - CAL 0: H: 21.059 V: 6.423
 - CAL 2: H: 31.196 V: 6.335
 - CAL 14: H: 41.417 V: 6.395
 - CAL 11: H: 51.396 V: 6.274
 - CAL 13 H: 51.396 V: 15.766
 - CAL 9: H: 41.542 V: 15.943
 - CAL 19: H: 31.509 V: 15.943
 - CAL 8: H: 21.094 V: 16.233

Run Draft

- Run **<run number>** : **<beam momentum> GeV beam, collimators at <setting>, <YES/NO TARGET>, <purpose>**
 - beamfile: **<000>**
 - beam momentum: **<+3/-3/+5> GeV**
 - Trigger Condition: **<S0S1 / S0S1FS0FS1 / other>**

- Event rate from CESAR: approx **<value>** events/spill
- Trigger rate from TDAQ: approx **<value>** events/spill
- XCET44: pressure **<value>** bar, events **<value>**, gas **CO2**
- XCET48: pressure **<value>** bar, events **<value>**, gas **CO2**
- Run started: **<time>**, **<day>**
- Run finished: **<time>**, **<day>**
- Number of events: ****
- Target position: **<position / ->**
- Target material thickness: **<C, W, Fe / 30 mm>**
- DESY Table position: **
- **Observation:**

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This run is quite short in terms of total number of events, after we have started to the run we went inside to make some changes for MMs.

- Run **1787523716 : 4 GeV beam**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **150899** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
 - Run started: **24-Aug-2026 00:21:56**
 - Run finished: **24-Aug-2026 00:38**
 - Number of events: **20k?**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **3 cm C**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:** things look ok, only maybe some particles are hitting with 4 GeV indeed.

Again 3 cm Graphite but more data

- Run **1787526649 : 4 GeV beam**
 - beamfile: **<019>**

- beam momentum: **+4 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Event rate from CESAR: approx **150899** events/spill
- Trigger rate from TDAQ: approx **1.4k** events/spill
- XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
- XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
- Run started: **24-Aug-2026 01:10:49**
- Run finished: **** 24-Aug-2026 02:24:08****
- Number of events: **444741**
- Target position: **49 cm from the yellow**
- Target material thickness: **3 cm C**
- DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
- **Observation:** things look ok, only maybe some particles are hitting with 4 GeV indeed.

We are switchinf 3 cm Aluminium

- Run **1787532112 : 4 GeV beam**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **150899** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
 - Run started: ****24-Aug-2026 02:41:52 ****
 - Run finished: **** 24-Aug-2026 03:51:09 ****
 - Number of events: **459871**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **3 cm alu**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:** things look ok, only maybe some particles are hitting with 4 GeV indeed.

background run at 4 GeV

- Run **1787536596 : 4 GeV beam, background run**

- beamfile: **<019>**
- beam momentum: **+4 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Event rate from CESAR: approx **150899** events/spill
- Trigger rate from TDAQ: approx **1.4k** events/spill
- XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
- XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
- Run started: ****24-Aug-2026 03:56:36 ****
- Run finished: **** 24-Aug-2026 04:10:10 ****
- Number of events: **91274**
- Target position: **49 cm from the yellow**
- Target material thickness: **no target**
- DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
- **Observation:** bkg run, we hope timepix matches.

we will now switch to the 3 cm tungsten target.

- Run **1787538016 : 4 GeV beam, physics run 3 cm tungsten**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **150899** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
 - Run started: **24-Aug-2026 04:20:16**
 - Run finished: **** 24-Aug-2026 05:04:47****
 - Number of events: **300768**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **3 cm W**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:** phys run, 3 cm tungsten

we've now switched to 1.2 cm of copper.

- Run **1787541067 : 4 GeV beam, physics run 1.2 cm Cu**
 - beamfile: **<019>**

- beam momentum: **+4 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Event rate from CESAR: approx **150899** events/spill
- Trigger rate from TDAQ: approx **1.4k** events/spill
- XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
- XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
- Run started: **24-Aug-2026 05:11:07**
- Run finished: **24-Aug-2026 05:56:13**
- Number of events: **300463**
- Target position: **49 cm from the yellow**
- Target material thickness: **1.2 cm Cu**
- DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
- **Observation:** phys run, 1.2 cm Cu.

this should be the final run of the 4 GeV beam, we also have the bkg run. the remaining runs are 3 GeV bkg (increase the XCET48 to 2 bar), 3 GeV 5 cm graphite, 4 GeV 4 cm Alu, (increase the XCET48 to 4 bar and XCET44 to 8-9 bar.) 1.8 GeV bkg run. We also actually need a run that measures the cherenkovs at the same pressure. I believe we cannot do this one.

- Run **1787544111** : **3 GeV beam,background run no target, **
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **112391** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **82539**, gas **CO2**
 - XCET48: pressure **1.991** bar, events **6086**, gas **CO2**
 - Run started: **24-Aug-2026 06:01:51**
 - Run finished: **24-Aug-2026 06:21:58**
 - Number of events: **77534**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **BACKGROUND RUN FOR 3 GeV**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:** aLL OKAY

We are switcing to 3 GeV runs to complete data set 4 cm ALU

- Run **1787545748** : **3 GeV beam, 4m alu run, **
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **117706** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **84932**, gas **CO2**
 - XCET48: pressure **1.991** bar, events **6416**, gas **CO2**
 - Run started: **24-Aug-2026 07:20:55**
 - Run finished: ** *24-Aug-2026 *
 - Number of events: ****
 - Target position: **49 cm from the yellow**
 - Target material thickness: **4 CM GRAPHITE**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation**: aLL OKAY
- Run **1787545748** : **3 GeV beam, 4Cm GRAPHITE, **
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **112391** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **82539**, gas **CO2**
 - XCET48: pressure **1.991** bar, events **6086**, gas **CO2**
 - Run started: **24-Aug-2026 06:29:08**
 - Run finished: ** 24-Aug-2026 08:12:00
 - Number of events: **117800**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **4 CM ALU**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation**: aLL OKAY

1787552171 1.8 gEv BKG RUN WITH THE COLLIMATORS AT ± 50 , AND CHERENKOVs AT 9 AND 4 BAR.

24-Aug-2026 08:16:11

24-Aug-2026 09:35:04 23645

END OF TESTBEAMMM

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after fondue, we have finished the calibration runs.

- Run **1787444130** : **1.8 GeV beam, collimator XCHV011 used to be at -8 but we have it at 50.036 and XCHV012 USED TO BE AT -8 BUT WE HAVE IT AT 50.052, XCHV014 USED TO BE AT -8 BUT IS AT 50.017, XCHV015 used to be at -8 but is at 50.084, XCHV027 used to be at -20 but is at 49.987 , 1cm Graphite/c target, purpose: PHYSICS RUNS**

- beamfile: **018**
- beam momentum: **1.8 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Event rate from CESAR: approx **19180** events/spill
- Trigger rate from TDAQ: approx **150** events/spill
- XCET44: pressure **7.991** bar, events **13347**, gas **CO2**
- XCET48: pressure **3.991** bar, events **1704**, gas **CO2**
- Run started: ****
- Run finished: ****
- Number of events: ****
- Target position: **40 cm from ecal**
- Target material thickness: **Graphite 1cm**
- DESY Table position: H: 35.170 V:1.276**
- **Observation:** it is just so so so so so low.

- Run **1787473551** : **1.8 GeV beam, collimator XCHV011 used to be at -8 but we have it at 50.036 and XCHV012 USED TO BE AT -8 BUT WE HAVE IT AT 50.052, XCHV014 USED TO BE AT -8 BUT IS AT 50.017, XCHV015 used to be at -50.263 but is at 50.084, XCHV027 used to be at -20 but is at 49.987 , 1cm Al target, purpose: PHYSICS RUNS**

- beamfile: **018**
- beam momentum: **1.8 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Event rate from CESAR: approx **18385** events/spill
- Trigger rate from TDAQ: approx **160** events/spill
- XCET44: pressure **7.991** bar, events **12788**, gas **CO2**
- XCET48: pressure **3.991** bar, events **1626**, gas **CO2**
- Run started: **23-Aug-2026 10:25:51**

- Run finished: **13:32:15**
- Number of events: **71808**
- Target position: **40 cm from ecal**
- Target material thickness: **Aluminium 1cm**
- DESY Table position: H: 35.170 V:1.276**
- **Observation:** we don't know, everything is working. Asta needs a bit of time with the beam, we turned on the smol cute run again, the beamfile is 1787484825 ,start time : 23-Aug-2026 13:33:45, stop time: 23-Aug-2026 13:54:17 and event count: 3649.

Now we want to do the 4 GeV runs, for this, for the alignment, we will go to the center of cal14, we will also lower the cherenkovs.

The changes that we made till now :D

From	To	New Length	Previous len
Below fan (left BP)	Cal 19	47.3	47.1
Top of fan	Cal 02	47.1	46.7
BP right top of the box	Cal 07	46.3	46.1
Katherine box DS	Cal 11	40.1	39.9
Top of time pix	Cal 14	49.3	49.3
Target	Cal 02	50.1	49.9
Timepix DS	Yellow slab DS	63.32	63.15

Now what we did inside :D

We aligned the center of the timepix to good precision (maybe :D) and then we started the run but forgot to align the FS but as a good experimentalist (yup) we did completed that too :D and then we started our run :D

- Run **1787489827 : 4 GeV beam, 2 cm W target , purpose:PHYSICS RUNS**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **66564** events/spill
 - Trigger rate from TDAQ: approx **1k** events/spill
 - XCET44: pressure **7.998** bar, events **46584**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **1204**, gas **CO2**
 - Run started: **23-Aug-2026 14:57:07**

- Run finished: **23-Aug-2026 16:25:15**
- Number of events: **221352**
- Target position: **49 cm from the yellow**
- Target material thickness: **2 cm W**
- DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
- **Observation:**we stopped at 200k.

We changed the target to 2.42 cm of copper in the beam area

- Run **1787495722 : 4 GeV beam, 2.2 cm copper target , purpose:PHYSICS RUNS**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **66564** events/spill
 - Trigger rate from TDAQ: approx **1k** events/spill
 - XCET44: pressure **7.998** bar, events **46584**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **1204**, gas **CO2**
 - Run started: **23-Aug-2026 16:35:22**
 - Run finished: **23-Aug-2026 17:59:30**
 - Number of events: **216422**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **2.4 cm Cu**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:**we stopped at 216k.

we've changed the target to 1 cm of alu, which should help both teams we hope, for the team attopion since the stats are a bit more important we will wait a bit lon on this one.

- Run **1787502393 : 4 GeV beam, 1 cm aluminium target , purpose:PHYSICS RUNS**
we've hopened the collimators for more particles, collimator numbers are 11 12 14 15, and they were at ± 20 , now they are at ± 40 .
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **150899** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **3092**, gas **CO2**

- Run started: **23-Aug-2026 18:26:33**
- Run finished: **23-Aug-2026 20:45:04**
- Number of events: **502489**
- Target position: **49 cm from the yellow**
- Target material thickness: **1 cm Al**
- DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
- **Observation:**things look ok, only maybe some particles are hitting with 4 GeV indeed.

we will switch to 1 cm graphite.

- Run **1787514996 : 4 GeV beam**
 - beamfile: **<019>**
 - beam momentum: **+4 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **150899** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **7.998** bar, events **107130**, gas **CO2**
 - XCET48: pressure **1.103** bar, events **3092**, gas **CO2**
 - Run started: **23-Aug-2026 21:56:36**
 - Run finished: **24-Aug-2026 00:13:45**
 - Number of events: **500218**
 - Target position: **49 cm from the yellow**
 - Target material thickness: **1 cm C**
 - DESY Table position: *H: 35.165, V:1.278 ###center of the calorimeters.*
 - **Observation:**things look ok, only maybe some particles are hitting with 4 GeV indeed.

We will switch to 3 cm Graphite target @4 GeV

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- Run **1787368460 : 3 GeV beam, 1.2 cm Cu target , purpose:PHYSICS RUNS**
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**

- XCET44: pressure **2.0** bar, events **5110**, gas **CO2**
- XCET48: pressure **4.0** bar, events **88488**, gas **CO2**
- Run started: **22-Aug-2026 05:14:20**
- Run finished: **22-Aug-2026 10:10:18**
- Number of events: **610399**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **1.2 cm Copper**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** We stopped and restarted the run. We had no beam so we checked the HVs, but there wasn't an issue with them. Later we took the beam stopper off, which fixed the beam. Timepix didn't record till around 10:00:00.

When we came to the control center the Timepix wasn't working. There was a communication issue. After fixing the computer, we reseted the Timepix by unplugging it and counting to 10. After plugging it back we checked the PC again, but the issue wasn't fixed. So we closed the firewall which fixed the issue.

Close the firewall after restarting the PC for Timepix to work.

- Run **1787387158 : 3 GeV beam, 1.2 cm Cu target , purpose:PHYSICS RUNS**
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0** bar, events **4981**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **91412**, gas **CO2**
 - Run started: **22-Aug-2026 10:25:58**
 - Run finished: **22-Aug-2026 10:56:11**
 - Number of events: **61340**
 - Target position: **45cm from the bosch profiles**

- Target material thickness: **1.2 cm Copper**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*

Changed the target from 1.2 cm Copper to 2.4 cm Copper. We wanted 3 cms, but we didn't have enough Copper targets to do so.

- Run **1787390563 : 3 GeV beam, 2.4 cm Cu target , purpose:PHYSICS RUNS**

- beamfile: **<008>**
- beam momentum: **+3 GeV**
- Trigger Condition: **S0S1FS0FS1**
- XCET44: pressure **2.0** bar, events **5110**, gas **CO2**
- XCET48: pressure **4.0** bar, events **92486**, gas **CO2**
- Run started: **22-Aug-2026 11:22:43**
- Run finished: **22-Aug-2026 14:22:12**
- Number of events: **402046**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **2.4 cm Copper**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*

observation: looks okay!Due to QDC change calorimeters must be realigned and re-calibrated

REALIGNMENT AND RE-CALIBRATION OF THE CALORIMETERS

DESY TABLE MOVEMENTS/LOCATIONS:

CAL 18

- Horizontal = 41.405
- Vertical = -3.650

CAL 7

- Horizontal = 51.384
- Vertical = -3.705

CAL 4

- Horizontal = 29.195
- Vertical = -3.617

CAL 17

- Horizontal = 19.074
- Vertical = -3.437

We have the center locations of CAL 18, 7, 4, 17. Batteries in laser needs to be replaced. So we will get the centers of the other calorimeters later. We will proceed with calibration runs for the 4 calorimeters now. We have decided to re-do the calibration for all the ECals but only for 1 energy, 2GeV

- Run **1787407961** : **2 GeV electron beam, no target , purpose:Calibration Run for CAL 17**
- beamfile: **<001>**
- beam momentum: **2 GeV**
- Trigger Condition: **S0S1**
- XCET44: pressure **2.0** bar, events **5575**, gas **CO2**
- XCET48: pressure **4.0** bar, events **5751**, gas **CO2**
- Run started: **22-Aug-2026 16:12:41**
- Run finished: **22-Aug-2026 16:18:48**
- Number of events: **22034**
- DESY Table position: *H: 19.074, V:-3.437 ###center of the calorimeter 17.*

observation: It was centered at 2200, and it made us scared if it hits to the maximum value at 3 GeV. that's why we will tke 3 spill run with 3 gev beamfile.

- Run ****1787408396**** : ****3 GeV electron beam, no target , purpose:Calibration R**



- beamfile: **<002>**
- beam momentum: **3 GeV**
- Trigger Condition: **S0S1**

- XCET44: pressure **2.0** bar, events **6107**, gas **CO2**
- XCET48: pressure **4.0** bar, events **6445**, gas **CO2**
- Run started: **22-Aug-2026 16:19:56**
- Run finished: **22-Aug-2026 16:21:38**
- Number of events: **7051**
- DESY Table position: *H: 19.074, V:-3.437 ###center of the calorimeter 17.*
observation: It looked okay, not going beyond 4000.

Continuing with the calibration runs.

- Run ****1787408731**** : ****2 GeV electron beam, no target , purpose:Calibration Run**

- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****
- XCET44: pressure ****2.0** bar, events ****5345****, gas ****C02******
- XCET48: pressure ****4.0** bar, events ****5388****, gas ****C02******
- Run started: ****22-Aug-2026 16:25:31****
- Run finished: ****22-Aug-2026 16:31:30****
- Number of events: ****20372****
- DESY Table position: *H: 29.196, V:-3.616 ###center of the calorimeter 4.*

observation: It looked okay, centered around 2100.

- Run ****1787409242**** : ****2 GeV electron beam, no target , purpose:Calibration Run fo**

- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****
- XCET44: pressure ****2.0** bar, events ****5345****, gas ****C02******
- XCET48: pressure ****4.0** bar, events ****5388****, gas ****C02******
- Run started: ****22-Aug-2026 16:34:02****
- Run finished: ****22-Aug-2026 16:40:29****
- Number of events: ****21964****
- DESY Table position: *H: 51.386, V:-3.704 ###center of the calorimeter 7.*

observation: Looked okay centered around 2000.

- Run ****1787409712**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**

- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****
- XCET44: pressure ****2.0** bar, events ****5345****, gas ****C02******
- XCET48: pressure ****4.0** bar, events ****5388****, gas ****C02******
- Run started: ****22-Aug-2026 16:41:52****
- Run finished: ****22-Aug-2026 16:47:33****
- Number of events: ****20382****
- DESY Table position: *H: 41.404, V:-3.652 ###center of the calorimeter 18.*

- Observation: cal14 is centered at 1800.

Going in for getting the centers of the other calorimeters.



CAL 14

- Horizontal = 41.394
- Vertical = 6.108

CAL 11

- Horizontal = 50.994
- Vertical = 6.108

CAL 2

- Horizontal = 29.071
- Vertical = 6.102

CAL 0

- Horizontal = 19.050
- Vertical = 6.189

CAL 8

- Horizontal = 19.218
- Vertical = 16.065

CAL 19

- Horizontal = 29.326
- Vertical = 16.036

CAL 9

- Horizontal = 41.407
- Vertical = 15.991

CAL 13

- Horizontal = 51.343
- Vertical = 15.833

CAL 12

- Horizontal = 52.037
- Vertical = -13.346

CAL 1

- Horizontal = 42.162
- Vertical = -13.261

CAL 5

- Horizontal = 29.727
- Vertical = -13.126

CAL 10

- Horizontal = 19.601
- Vertical = -13.057

we will continue will cal10.

- Run ****1787414494**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****

- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:01:34****
- Run finished: ****22-Aug-2026 18:08:34****
- Number of events: ****25352****
- DESY Table position: ***H: 19.601, V:-13.057 ###center of the calorimeter 10.***
- Observation: cal10 is centered at 1700.

- Run ****1787414978**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****

- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:09:38****
- Run finished: ****22-Aug-2026 18:12****
- Number of events: ****x****
- DESY Table position: ***H: 28.726, V:-13.122 ###center of the calorimeter 5.***
- Observation: cal5 is centered at 2100.

- Run ****1787415228**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****

- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:13:48****
- Run finished: ****22-Aug-2026 18:19:18****
- Number of events: ****20211****
- DESY Table position: ***H: 29.730, V:-13.122 ###center of the calorimeter 5.***
- Observation: cal5 is centered at 2100.

- Run ****1787415623**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****

- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:20:23****
- Run finished: ****22-Aug-2026 18:26:19****
- Number of events: ****22000****
- DESY Table position: ***H: 42.163, V:-13.257 ###center of the calorimeter 1.***
- Observation: cal1 is centered at 1600.

- Run ****1787416044**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****
-
- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:27:24****
- Run finished: ****22-Aug-2026 18:33:32****
- Number of events: ****x****
- DESY Table position: ***H: 52.036, V:-13.346 ###center of the calorimeter 12.***
- Observation: **cal12 is centered at 1200.**
-
- Run ****1787416522**** : ****2 GeV electron beam, no target , purpose:Calibration Run for**
- beamfile: ****<001>****
- beam momentum: ****2** GeV**
- Trigger Condition: ****S0S1****
-
- XCET44: pressure ****2.0** bar, events **5345**, gas **C02****
- XCET48: pressure ****4.0** bar, events **5388**, gas **C02****
- Run started: ****22-Aug-2026 18:35:22****
- Run finished: ****22-Aug-2026 18:40:43****
- Number of events: ****20209****
- DESY Table position: ***H: 51.348, V:15.831 ###center of the calorimeter 13.***
- Observation: **cal13 is centered at 1700.**

we will leave with graphite 5 cm at 3 gev pos hadron beam.

change of plans, 5 cm graphite doesn't fit in between timepix

- Run **1787417290 : 3 GeV pos hadron beam, 1 cm alu target , purpose:missing half of the 1 cm alu with timepix**
 - beamfile: **<008>**
 - beam momentum: **3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0 bar, events x, gas CO2**
 - XCET48: pressure **4.0 bar, events x, gas CO2**
 - Run started: **22-Aug-2026 18:48:10**
 - Run finished: **22-Aug-2026 22:10:18**

- Number of events: **500k**
- DESY Table position: *H: 35.164, V:1.278 ###center of the calorimeters.*
- Observation: no no tworworking cal or not working timepix.

we will continue with the calib runs.

- Run **1787430082 : 2 GeV electron beam, no target , purpose:Calibration Run for CAL 9**
 - beamfile: **<001>**
 - beam momentum: **2 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5171**
 - XCET44: pressure **2.0** bar, events **5018**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5036**, gas **CO2**
 - Run started: **22-Aug-2026 22:21:22**
 - Run finished: **22-Aug-2026 22:27:44**
 - Number of events: **20209**
 - DESY Table position: *H: 41.410, V:15.995 ###center of the calorimeter 9.*
 - Observation: cal9 is centered at 2400. it is too much, we might take super short run with the 3 GeV.
- Run **1787430653 : 3 GeV electron beam, no target , purpose:Calibration Run for CAL 9**
 - beamfile: **<002>**
 - beam momentum: **3 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5837**
 - XCET44: pressure **2.0** bar, events **5521**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5797**, gas **CO2**
 - Run started: **22-Aug-2026 22:30:53**
 - Run finished: **22-Aug-2026 22:36:24*
 - Number of events: **20229**
 - DESY Table position: *H: 41.410, V:15.995 ###center of the calorimeter 9.*
 - Observation: cal9 is centered at 3500 count. it is at the limit.

- Run **1787431119 : 2 GeV electron beam, no target , purpose:Calibration Run for CAL 19**
 - beamfile: **<001>**
 - beam momentum: **2 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 22:38:39**
 - Run finished: **22-Aug-2026 22:44:35**
 - Number of events: **21415**
 - DESY Table position: *H: 29.327, V:16.035 ###center of the calorimeter 19.*
 - Observation: cal19 is centered at 1200 ADC count, mais not much to do, HV is at 1474V.
- Run **1787431583 : 2 GeV electron beam, no target , purpose:Calibration Run for CAL 8**
 - beamfile: **<001>**
 - beam momentum: **2 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 22:46:23**
 - Run finished: **22-Aug-2026 22:52:20**
 - Number of events: **21278**
 - DESY Table position: *H: 19.214, V:16.068 ###center of the calorimeter 8.*
 - Observation: cal8 is centered at 2200 QDC count.
- Run **1787432044 : 2 GeV electron beam, no target , purpose:Calibration Run for CAL 0**
 - beamfile: **<001>**
 - beam momentum: **2 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**

- XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
- XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
- Run started: **22-Aug-2026 22:54:04**
- Run finished: **22-Aug-2026 22:59:57**
- Number of events: **21178**
- DESY Table position: *H:19.050, V:6.187 ###center of the calorimeter 0.*
- Observation: cal0 is centered at 1900 QDC count.
- Run **1787432452** : **2 GeV electron beam, no target , purpose:Calibration Run for CAL 2**
 - beamfile: **<001>**
 - beam momentum: **2** GeV
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 23:00:52**
 - Run finished: **22-Aug-2026 23:07:08**
 - Number of events: **21186**
 - DESY Table position: *H:29.070, V:6.101 ###center of the calorimeter 2.*
 - Observation: cal2 is centered at 2200 QDC count.
- Run **1787432906** : **2 GeV electron beam, no target , purpose:Calibration Run for CAL 11**
 - beamfile: **<001>**
 - beam momentum: **2** GeV
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 23:08:26**
 - Run finished: **22-Aug-2026 23:14:24**
 - Number of events: **21225**
 - DESY Table position: *H:50.990, V:6.108 ###center of the calorimeter 11.*
 - Observation: cal11 is centered at 1800 QDC count.

- Run **1787433329 : 2 GeV electron beam, no target , purpose:Calibration Run for CAL 14**
 - beamfile: **<001>**
 - beam momentum: **2 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **1700**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 23:15:29**
 - Run finished: **22-Aug-2026 23:21:55**
 - Number of events: **x**
 - DESY Table position: *H:41.398, V:6.108 ###center of the calorimeter 14.*
 - Observation: cal14 is centered at 2200 QDC count.
- Run **1787433886 : 3 GeV electron beam, no target , purpose:Calibration Run for CAL 14**
 - beamfile: **<002>**
 - beam momentum: **3 GeV**
 - Trigger Condition: **S0S1**
 - Trigger Rate TDAQ: **120**
 - Trigger Rate CESAR: **5236**
 - XCET44: pressure **2.0** bar, events **5095**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **5120**, gas **CO2**
 - Run started: **22-Aug-2026 23:24:46**
 - Run finished: **22-Aug-2026 23:2**
 - Number of events: **52k**
 - DESY Table position: *H:41.398, V:6.108 ###center of the calorimeter 14.*
 - Observation: cal14 is centered at 3200 QDC count.

2026.08.21

we've arrived to the control room at 9:32. Calorimeters look fine (SHOCKING) but TimePix stopped at 344 reps, we have restarted again. We stopped the beam at 9:59, because of the social media team.

data recording is finished, however we have noticed that timepix is misaligned. Laser cross was still visible on TimePix chip but the vertical line was in a bit left side I did not think to take the picture :(

I believe while we are trying to take out the QDC we have moved the timepix :(so this run is also rubbish then... we will realign it again.

We have made the alignment, ready for background run. Here is the distances after our alignment.

from	to	distance (cm)
left BP	CAL8	46.7
right BP	CAL9	46.7
left BP	CAL10	44
right BP	CAL1	44.7
right BP	CAL 18`	45.2
left BP	BoDESY	41.7
right BP	BoDESY	41.5

- Run **1787269191** : **3 GeV beam, collimators at beam_ref, 1cm graphite , purpose: get data**
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0** bar,gas **CO2**
 - XCET48: pressure **4.0** bar,gas **CO2**
 - Run started: **01:39:51, 21-Aug-2026**
 - Run finished: **11:17:2829,21-Aug-2026**
 - Number of events: **2010193**
 - Target position: **45cm from the bosch profiles**
 - Target material thickness: **<C, / 10 mm>**

- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** graphite 1 cm data. when we arrived in the morning the calorimeters looked fine, but after we made an entrance ch6 cal8 looked a bit with overflow.

we will take another background run. cal8 ch6 looks bad.

- Run **1787306041** : **3 GeV beam, collimators at beam_ref, no target , purpose: get background**

- beamfile: **<008>**
- beam momentum: **+3 GeV**
- Trigger Condition: **S0S1FS0FS1**
- XCET44: pressure **2.0** bar,gas **CO2**
- XCET48: pressure **4.0** bar,gas **CO2**
- Run started: **11:56:00, 21-Aug-2026**
- Run finished: **12:24,21-Aug-2026**
- Number of events: **~30k**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **no**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** CAL8, channel 6 behaves wiggly woogly wuby.

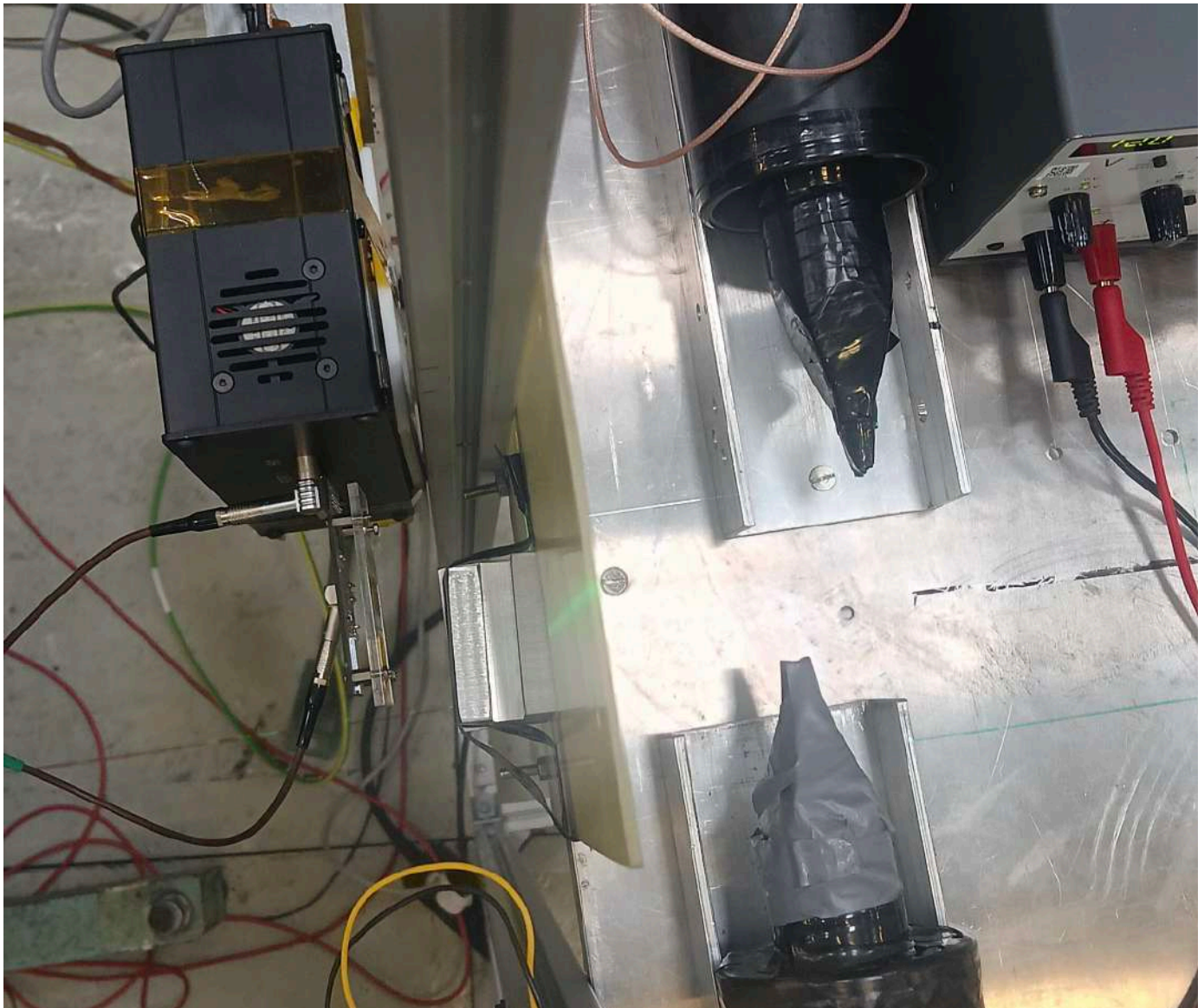
Markus has changed V792 at 12:25 with a new one, the serial number of current QDC V792 is: 97 (0x00000061). And now we will get a new short background run

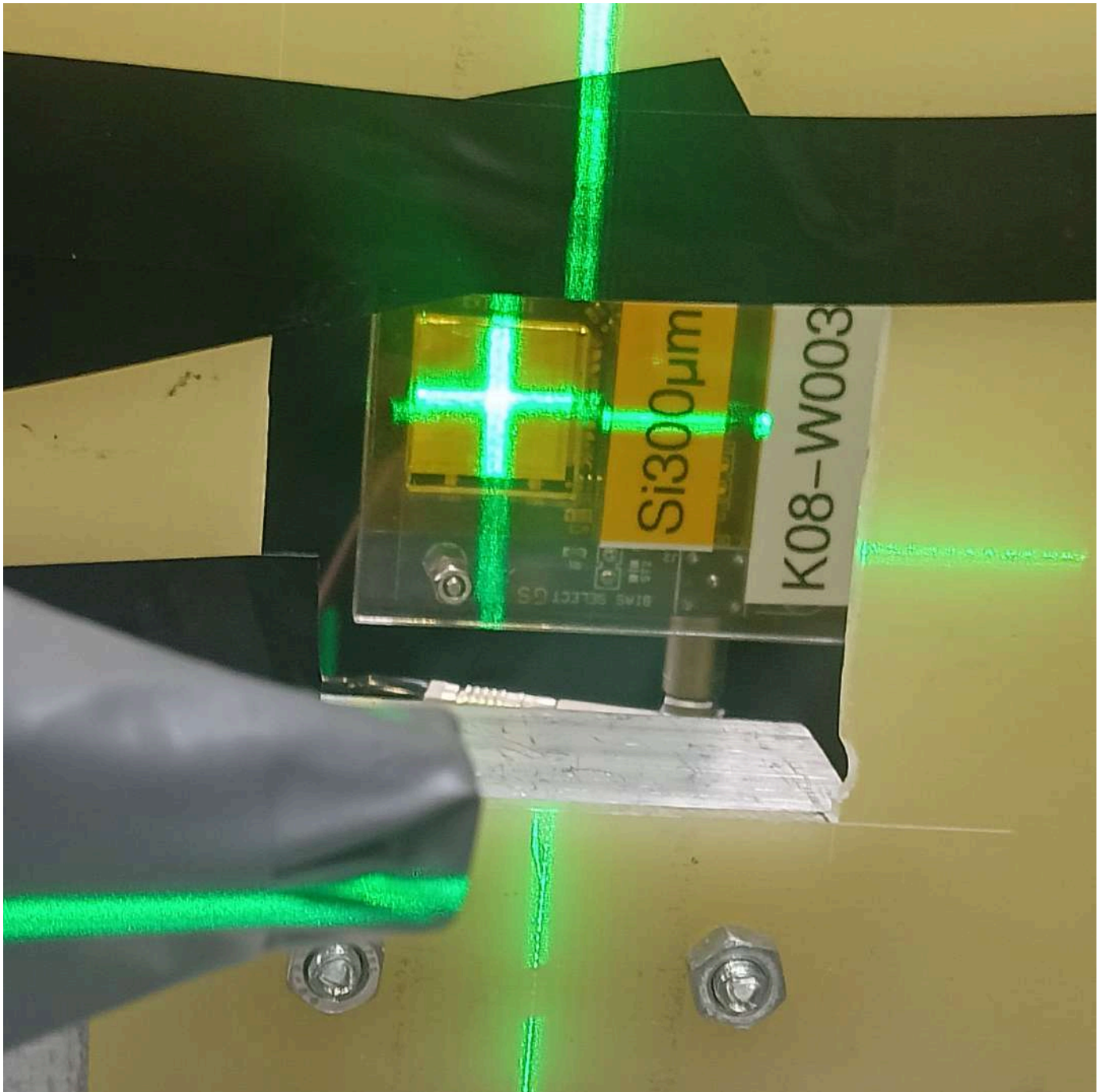
- Run **1787310616** : **3 GeV beam, collimators at beam_ref, no target , purpose: get background with new QDC board**

- beamfile: **<008>**
- beam momentum: **+3 GeV**
- Trigger Condition: **S0S1FS0FS1**
- XCET44: pressure **2.0** bar,gas **CO2**

- XCET48: pressure **4.0** bar, gas **CO2**
- Run started: **13:10:00, 21-Aug-2026**
- Run finished: **14:07:35, 21-Aug-2026**
- Number of events: **106424**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **no**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** looks okay for now. TimePix is also taking the data.

TIME TO ADD TARGEET-> 3.054 cm Aluminium @3GeV





- Run **1787316998** : **3 GeV beam**, collimators at beam_ref, Alu target ,
purpose:PHYSICS RUNS
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0 bar**,gas **CO2**

- XCET48: pressure **4.0** bar, gas **CO2**
- Run started: **14:56:38, 21-Aug-2026**
- Run finished: **17:00:25, 21-Aug-2026**
- Number of events: **231602**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **3 cm Aluminium**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:**
at 15:55 we have noticed the TimePix was not working, now we will have started again and it is working again.

Because of the DG visit we stoped the beam at 16:00. Ignore any data after that time.
(People might have changed the alignment)

We checked the alignments. Everything looks good.

- Run **1787325562 : 3 GeV beam, 3 cm Alu target , purpose:PHYSICS RUNS**
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0** bar, events **5110**, gas **CO2**
 - XCET48: pressure **4.0** bar, events **88488**, gas **CO2**
 - Run started: **17:19:22, 21-Aug-2026**
 - Run finished: **22:15:20, 21-Aug-2026**
 - Number of events: **1014948**
 - Target position: **45cm from the bosch profiles**
 - Target material thickness: **3 cm Aluminium**
 - DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
 - **Observation:**All looks FINE!

we've decided to get the data for the pedestal, this is with the beam on, HV OFF. Run number is 1787345503. Pedestal run start time 21-Aug-2026 22:51:43, run stop time: 21-Aug-2026 23:08:55, event size:53236.

- Run **1787346586 : 3 GeV beam, 1 cm Alu target , purpose:PHYSICS RUNS**

- beamfile: **<008>**
- beam momentum: **+3 GeV**
- Trigger Condition: **S0S1FS0FS1**
- XCET44: pressure **2.0** bar, events **5110**, gas **CO2**
- XCET48: pressure **4.0** bar, events **88488**, gas **CO2**
- Run started: **21-Aug-2026 23:09:46**
- Run finished: **04:52:59, 22-Aug-2026**
- Number of events: **1079724**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **1 cm Aluminium**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** All looks okay. TimePix seems stopped at 02:55.

We are switching to copper target!

Now time is : 18:00, so until Monday 08:00 we have 62 beam hours left. Let's think that we could not use 5 hours of this beamtime because of various reasons, we might have 58 hours. To collect 1m event we need to wait for 5,5 hours. We can have 10-11 long run before end of testbeam.

The RUNs COMPLETED

AttoPion	PionIST	Column 3
1 cm Alu	1 cm Alu	Text
3 cm Alu	3 cm Alu	

TO RUN's:

For AttoPion	For PionIST	Energy (GeV)
3 cm Alu		3
1 cm Alu		3
—	3 cm Cu	3
	1 cm Cu	3
1 cm W		1.5
3 cm W		1.5
	3 cm W	5
	1-2 cm C	5

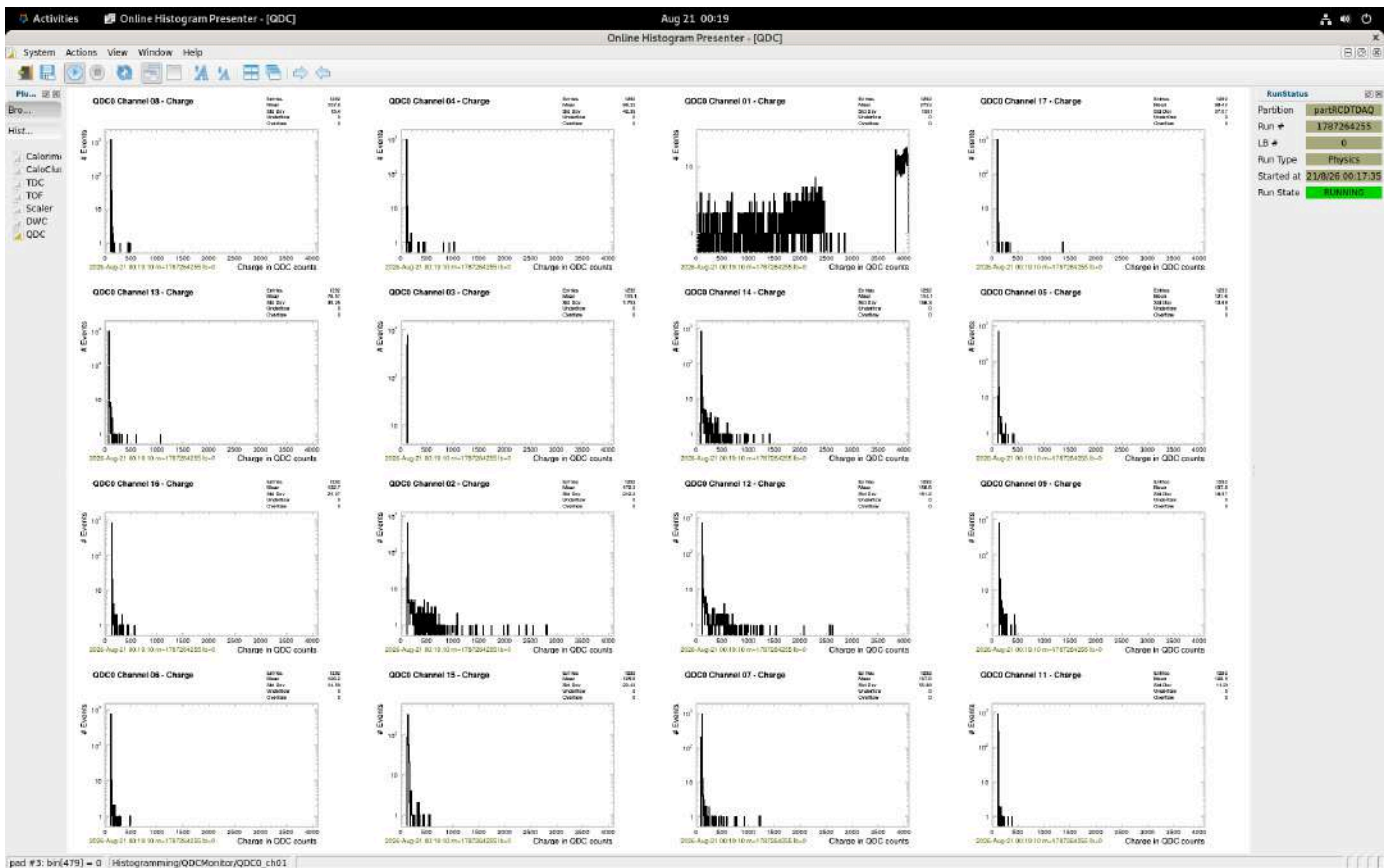
2026.08.20

We are still trying to find out the source of our problem with QDC channels/calorimeters.

Problematic channels from last night run: CHANNEL 01, CHANNEL 09, CHANNEL 13, CHANN

- Run **1787243607** : **3 GeV beam, collimators at beam_ref, 1cm graphite , purpose: TEST QDC Channel**

- beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **120421** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **2.007** bar, events **5824**, gas **CO2**
 - XCET48: pressure **3.993** bar, events **85826**, gas **CO2**
 - Run started: **18:33, 20-Aug-2026**
 - Run finished: **23:09,20-Aug-2026**
 - Number of events: **884719**
 - Target position: **45cm from the bosch profiles**
 - Target material thickness: **<C, / 10 mm>**
 - DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
 - **Observation:** channel 1 was sus it was showing woogly boogly graph
- :(



- Run **1787261776** : **3 GeV beam, collimators at beam_ref, 1cm graphite , purpose: we unplugged the cable of ch1 to understand the reason for the sus graph **
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.007** bar, gas **CO2**

- XCET48: pressure **3.993** bar, gas **CO2**
- Run started: **23:36, 20-Aug-2026**
- Run finished: **23:57,20-Aug-2026**
- Number of events: **89929**
- Target position: **45cm from the bosch profiles**
- Target material thickness: **<C, / 10 mm>**
- DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
- **Observation:** we ran this to find out what was the reason for the sus and unusual graph

this was a run to understand the problem of the Sus QDC graph we plugged out the ch1 cable and ran the beam again

(add graph)

- Run **1787264255 : 3 GeV beam, collimators at beam_ref, 1cm graphite , purpose: TEST QDC Channel again**
 - beamfile: **<008>**
 - beam momentum: **+3 GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - XCET44: pressure **2.0** bar,gas **CO2**
 - XCET48: pressure **4.0** bar,gas **CO2**
 - Run started: **00:17, 21-Aug-2026**
 - Run finished: **00:29,21-Aug-2026**
 - Number of events: **47991**
 - Target position: **45cm from the bosch profiles**
 - Target material thickness: **<C, / 10 mm>**
 - DESY Table position: *H: 35.167, V:1.278 ###center of the calorimeters.*
 - **Observation:** re-run to know the reason for the sus graph

2026.08.19

- Run **1787088655** : **3 GeV beam, TARGET: 1 cm of Tungsten!
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **106972** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78639**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5600**, gas **CO2**
 - Run started: **18-Aug-2026 23:30:55**
 - Run finished: **19-Aug-2026 10:08:28**
 - Number of events: **1482805**
 - DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
 - **Observation:** TimePix is recorded for 600 minutes, 10 hours, so it doesn't cover the entire run because we thought we would arrive earlier :D

CAL12 needs to be calibrated! (because we have change the qdc channel of it to qdc ch17.)

- Run **1787131925** : **3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate low Z material - Graphite(3cm) with TimePix.**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **107897** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78658**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5658**, gas **CO2**
 - Run started: **18-Aug-2026 11:32**
 - Run finished: ** 18-Aug-2026 14:00:45**
 - Number of events: **480878**
 - DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
 - CAL Gap Distance: ~3 cm

- **Observation:** over here we are having problem with RCDApp! We've decided to stop the run and restart the system to see if they will disappear. TimePix is continuing to have the data, it has GPIO1 and GPIO2 connected, GPIO1 is connected to EVTTRG
- Run **1787143237 : 3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate low Z material - Graphite(3cm) with TimePix.**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **107897** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78658**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5658**, gas **CO2**
 - Run started: **19-Aug-2026 14:40:37**
 - Run finished: **** 19-Aug-2026 21:16:04****
 - Number of events: **1334353**
 - DESY Table position: **** H: 35.167, V:1.278 ###center of the calorimeters.**
 - CAL Gap Distance: **~3 cm**
 - **Observation:** everything seems to be working out great (knock on wood) The TimePix data that covers this information is 757 to 1152. We will put tungsten at 3 cm just to see that probably we cannot build the pi0s :/.
- Run **1787168532 : 3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate low Z material - tungsten(3cm) with TimePix.**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **107897** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78658**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5658**, gas **CO2**
 - Run started: **19-Aug-2026 21:42**

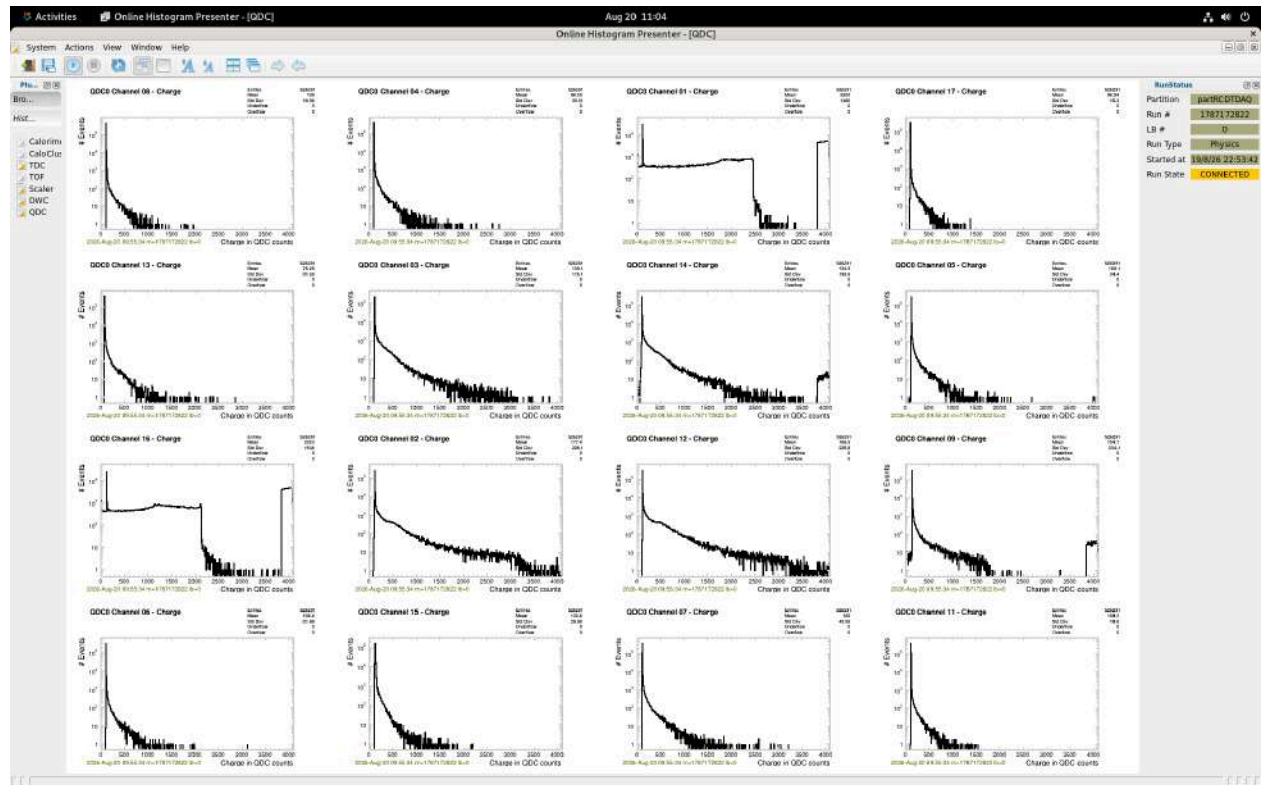
- Run finished: ** 19-Aug-2026 22:38**
- Number of events: **210780**
- DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
- CAL Gap Distance: ~3 cm
- **Observation:** everything looks okay! the TimePix data that covers this data is from 1174 to 1219. It is shorter than the data. The Timepix monitor become so busy after the 3 cm tungsten.

we will goo see theee moooon

we saw it, it was so beautifullll

- Run **1787172822 : 3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate low Z material - GGraphite(1cm) with TimePix.**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **107897** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78658**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5658**, gas **CO2**
 - Run started: **19-Aug-2026 22:53:42 **
 - Run finished: ** 20-Aug-2026 09:55:54 **
 - Number of events: ** 09:55:54**
 - DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
 - CAL Gap Distance: ~3 cm
 - **Observation:** The TimePix data that covers this run starts with the 1220. Unfortunately it seems TimePix stopped recording separate txt files 1313 at 00:34. But that files seems huge, we are not sure if it is corrupted or not. Also bad news on

the calorimeters.



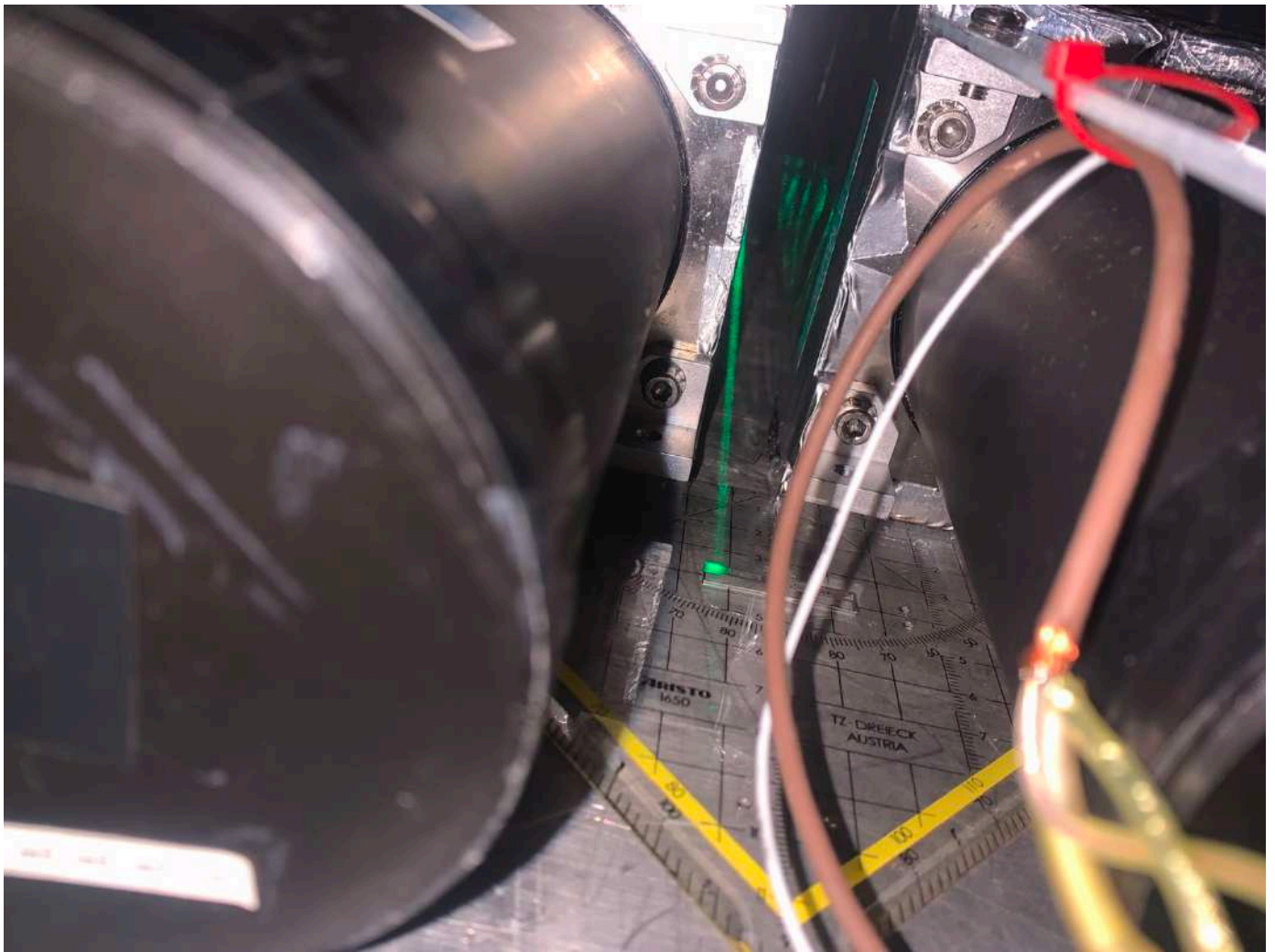
ch1 and ch16 is definitely problematic. Interesting part is that it is the same calorimeter, even at the new channel. We've went to talk to mustafa, he has suggested it looks like a memory issue, he asked if we reset in between. We will share the code with him, and we've got new qdc to test. Sad! We'll copy this event to next day too.

we are aiming to start a long run with 1 cm C and go home to sleeep

2026.08.18

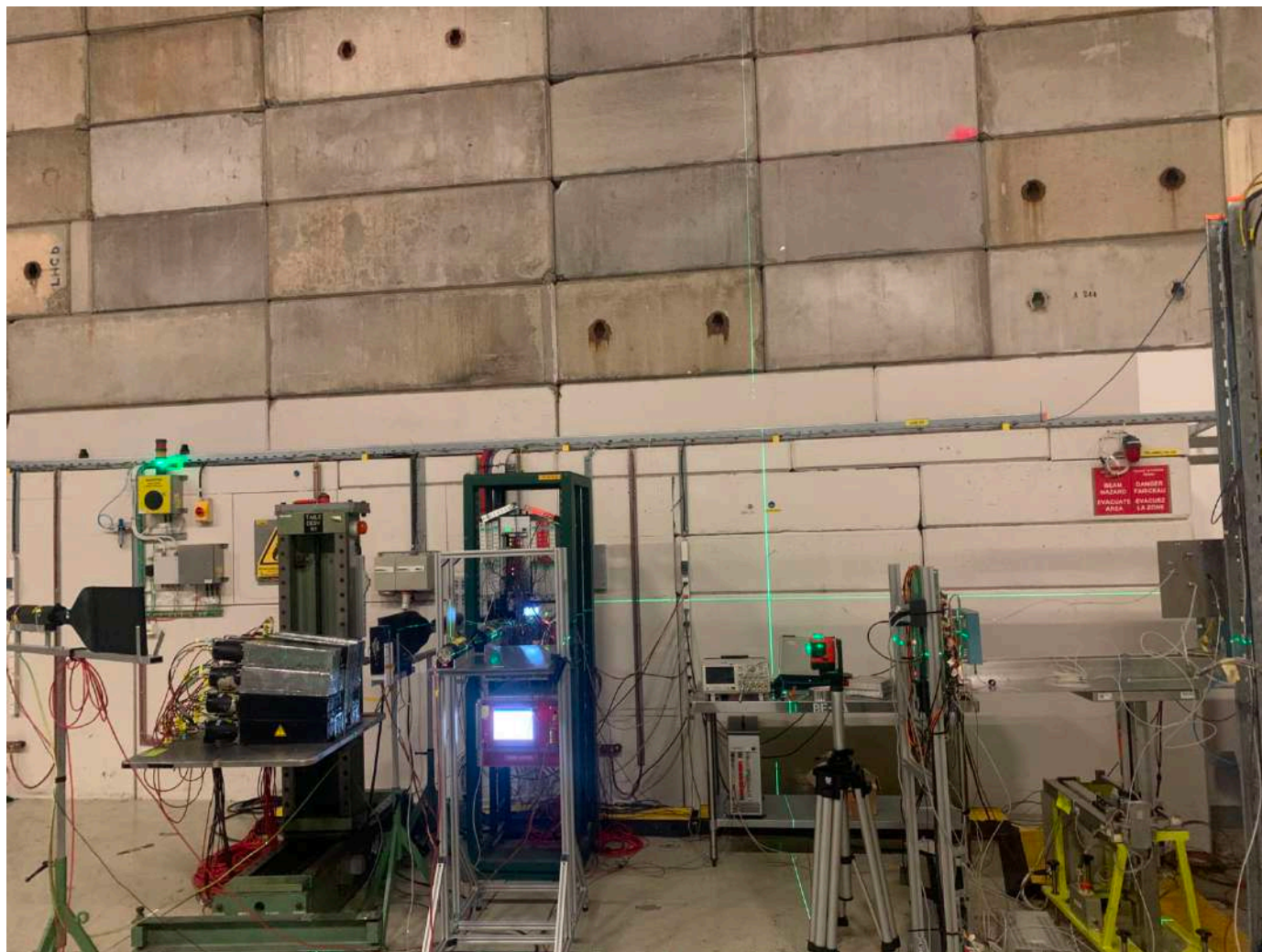
→Continuation from last night:

- We have corrected alignment of FS0, FS1 and center of CALs and their distance between each other. In previous positioning of CALs, laser were passing through the CAL14 and CAL18 but not the center of the gap. So, we have changed the CAL positions and then DESY table position as well.
- We have marked positions of FS PMT base on the bosch profiles.
- We have marked the wall vertically and wrote BL4S 2026.





- We have placed the TimePix in front of the beam and got 3 spills. There is no event on TimePix, so we will try again tomorrow. We (Aniket, Abhay, Berare and Seyma) are so tired to debug this thing now and also not 100% sure if the Katherine box and TimePix radiation hard or not. Needs to be check with TimePix people.



We have placed to 3 cm C target and we have ~3 cm gap between CALs.

- Run **1787004129** : **3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: 3 cm C target with ~2.8 - 3 cm gap between CALs.**
- beamfile: **<008>**
 - beam momentum: **<+3> GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **107907** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **3.994** bar, events **79054**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **5822**, gas **CO2**
 - Run started: **00:02, 18.08.2026**

- Run finished: **10:02, 18.08.2026**
- Number of events: **1539159**
- Target position: **44 & 45 cm bosh profile**
- Target material thickness: **C / 30 mm**
- DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
- CAL Gap Distance: 3 cm
- **Observation:**

We removed the target to measure the background.

- Run **1787045879** : **3 GeV beam, collimators at beam_ref, TARGET; NO, <GOAL: BACKGROUND RUN - NO TARGET, No TimePix**
- beamfile: **<008>**
 - beam momentum: **<+3> GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **102651** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **3.994** bar, events **74531**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **5149**, gas **CO2**
 - Run started: **11:37, 18.08.2026**
 - Run finished: **11:57, 18.08.2026**
 - Number of events: **50507**
 - DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
 - CAL Gap Distance: 3 cm
 - **Observation:** LOOKS NORMAL. CAL2 AND CAL12 HAS HIGHER BACKGROUND ~4K COUNTS, AS EXPECTED.
- Run **1787049075** : **3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate higher Z material - Tungsten(2cm) **
- beamfile: **<008>**
 - beam momentum: **<+3> GeV**
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **108025** vents/spill

- Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **3.994** bar, events **78279**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **5471**, gas **CO2**
 - Run started: **12:31, 18.08.2026**
 - Run finished: ****13:47,18.08.2026 ****
 - Number of events: **274068**
 - DESY Table position: **** H: 35.167, V:1.278 ###center of the calorimeters.**
 - CAL Gap Distance: 3 cm
 - **Observation:** stopped the run because CH10 QDC graph is suspicious

It is doing the same behaviour as seen earlier in the QDC log graph

We changed the channel 10 for calorimeter 12 to channel 16 and it seems to work.

From	To	Distance
Bottom of CAL9	Bottom of CAL19	21.3 mm
Top of CAL9	Top of CAL19	22.0 mm
CAL2	CAL14	23.5 mm
CAL4	CAL18	24.4 mm
CAL5	CAL1	24.9 mm
Top of CAL5	Top of CAL1	25.8 mm

- Run **1787058277** : ****3** GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: To investigate higher Z material - Tungsten(2cm) **
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **106972** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **77952**, gas **CO2**

- XCET48: pressure **2.002** bar, events **5600**, gas **CO2**
- Run started: **18-Aug-2026 15:04:37**
- Run finished: **15:04:37 20:50**
- Number of events: **744709**
- DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
- CAL Gap Distance: 3 cm
- **Observation:** no not working cal.

we have decided to take the target off and record a data in TDAQ and TimePix with the repeater method. We decided for 15 minutes of data taking, we will first start the TimePix and then TDAQ, and then we will give it to our poor students to correlate, good luck. first spill without tdaq. and we will also get the second SMA to BNC our goal is to also timestsamp the EVTTRG not just the CORBO we hope it will be helpful with the timeline analysis. Later update, we couldn't find SMA.

- Run **1787081424** : **3 GeV beam, collimators at beam_ref, TARGET; NO, <GOAL: to get background run with also TimePix to see if we can correlate the and to see how does the 3 GeV pos hadron beam looks with the TimePix.**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **106972** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **4.001** bar, events **78639**, gas **CO2**
 - XCET48: pressure **2.002** bar, events **5600**, gas **CO2**
 - Run started: **18-Aug-2026 21:30:24**
 - Run finished: **18-Aug-2026 21:53:30**
 - Number of events: **69157**
 - DESY Table position: ** H: 35.167, V:1.278 ###center of the calorimeters.
 - CAL Gap Distance: 3 cm
 - **Observation:** no not working cal.

- **TimePix** from measurement_9 file to measurement_23 file there it belongs to this data without any target and with 3 GeV pos hadron beam. Total timestamps on the TimePix is 40095, less than the TDAQ of course, because TimePix was set to record for 15 minutes. **Important note** over here we only had GPIO2, and it was recording the CORBO on the falling edge.

we are trying to move the TimePix close to the target as much as possible. And we would like to start 1 cm tungsten target, however we are a bit scared in terms of temperature of the TimePix, it can be monitored on the TrackLab but we don't know if it has protection system. That is why we are also looking for some fan in the lab, hope to find out some solution.

we've findout a 12V fan, it looks good enough for leaving over night now.

2026.08.17

Continuation from last night:

- Run **1786914869** : **3 GeV beam, collimators at beam_ref, TARGET; YES, <GOAL: repeating run from June BUT with 3 cm target**
- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **111035** vents/spill
 - Trigger rate from TDAQ: approx **<1200>** events/spill
- XCET44: pressure **3.994** bar, events **78639**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **5887**, gas **CO2**
 - Run started: **23:14:29, 16.08.2026**
 - Run finished: ****10:07:39,17.08.2026 ****
 - Number of events: **2011309**
 - Target position: **50 cm**
 - Target material thickness: **C / 30 mm**
 - DESY Table position: **** H: 36.151, V:1.180 ###center of the calorimeters.**
 - **Observation:** During this long run, Channel10, CAL12, created same weird shape like before with CAL8, CAL0 and CAL1. We will do the calibration for the cal12 again at a lower voltage, this was too high we think.

We are taking the measurement

From	To	Distance
DWC0 DS	DWC1 US	98.2 cm
DWC1 US	FS0 US	217.9 cm
FS0 US	FS1 US	3.24 cm
FS0 US	FS0 DS	0.7 cm
FS1 US	FS1 DS	0.6 cm
FS1 US	Target US	7.67 cm
Target US	Target DS	3 cm
Target DS	CAL2 US	46.3 cm

Changed the DESY tables position to (52.218, -13.079), and removed the target. Changed the beam to an electron beam. Changed the HV of Cal12 to 1500V and 445uA.

- Run ****1786960293**** : ******-3 GeV electron beam, collimators at beam_ref, TARGET; NO, <G



- beamfile: **<002>**
 - beam momentum: **<-3>** GeV
 - Trigger Condition: **S0S1**
 - Event rate from CESAR: approx **5221** vents/spill
 - Trigger rate from TDAQ: approx **1600** events/spill
 - XCET44: pressure **3.994** bar, events **5214**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **4987**, gas **CO2**
 - Run started: **17.08.2026, 11:51:33**
 - Run finished: **17.08.2026, 12:06:20**
 - Number of events: **47k**
 - DESY Table position: ****** H: 52.218, V:-13.079

- Run **1786961253** : **-2 GeV electron beam, collimators at beam_ref, TARGET; NO, <GOAL: Repeating our calibration runs for Cal12**
- beamfile: **<001>**
 - beam momentum: **<-2> GeV**
 - Trigger Condition: **S0S1**
 - Event rate from CESAR: approx **** vents/spill
 - Trigger rate from TDAQ: approx **1600** events/spill
 - XCET44: pressure **3.994** bar, events **5214**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **4987**, gas **CO2**
 - Run started: **17.08.2026, 12:06:20**
 - Run finished: **17.08.2026, 12:57:30**
 - Number of events: **204k**
 - DESY Table position: ** H: 52.218, V:-13.079
 - Run **1786964403** : **-1 GeV electron beam, collimators at beam_ref, TARGET; NO, <GOAL: Repeating our calibration runs for Cal12**
- beamfile: **<000>**
 - beam momentum: **<-1> GeV**
 - Trigger Condition: **S0S1**
 - Event rate from CESAR: approx **2166** vents/spill
 - Trigger rate from TDAQ: approx **1600** events/spill
 - XCET44: pressure **3.994** bar, events **2157**, gas **CO2**
 - XCET48: pressure **1.999** bar, events **2155**, gas **CO2**
 - Run started: **17.08.2026, 13:00:03**
 - Run finished: **17.08.2026, 13:38:12** :
 - Number of events: **111836**
 - DESY Table position: ** H: 52.218, V:-13.079

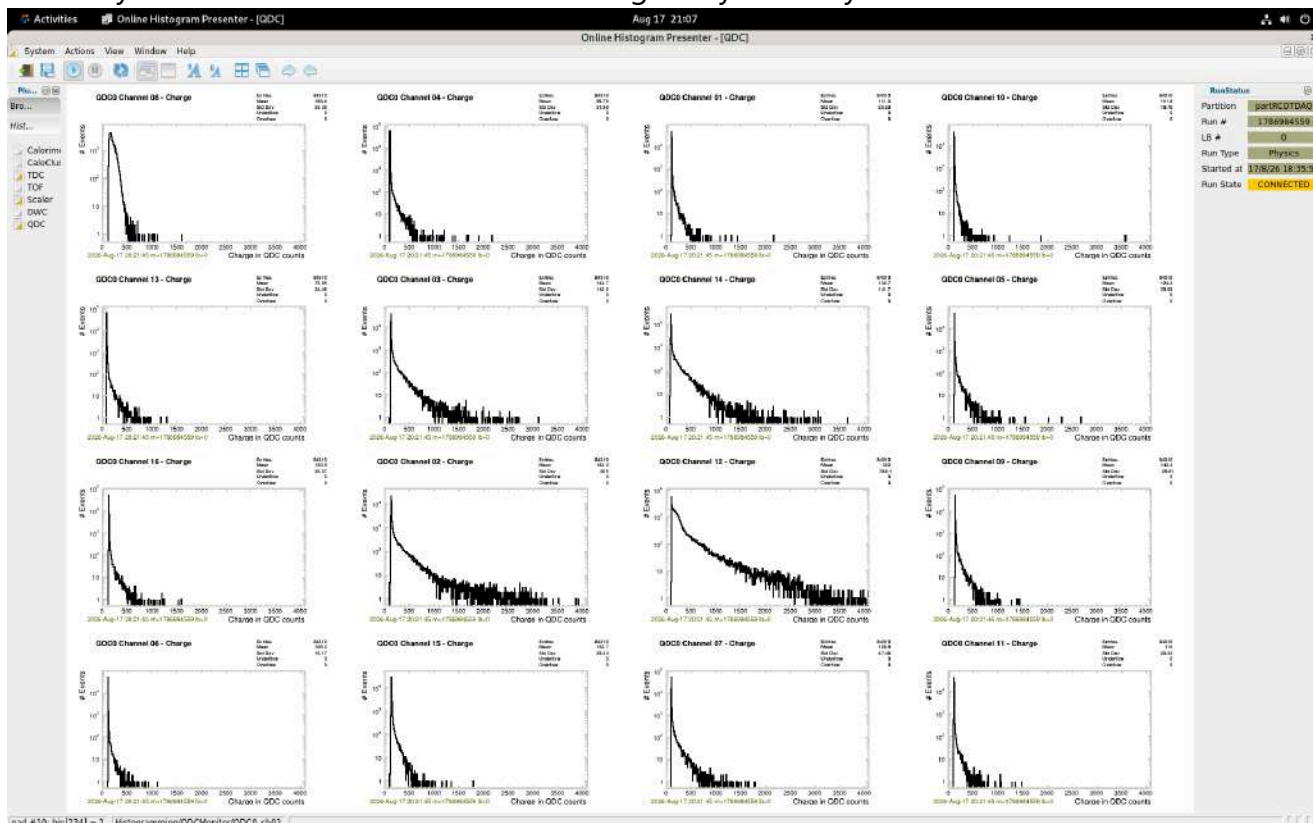
NOTE: We had issues with loading the beamfile although we tried it more than 4 times. And then, we called the PS control room, now we are waiting for our beamline physicist. It was because we have stayed inside so long that they took us out of the cycle :D
- Run **1786984559** : **+3 GeV positive hadron beam NO TARGET, background run **

- beamfile: **<008>**
 - beam momentum: **<+3>** GeV
 - Trigger Condition: **S0S1FS0FS1**
 - Event rate from CESAR: approx **5221** vents/spill
 - Trigger rate from TDAQ: approx **1200** events/spill

it seems a bit low than our previous value, the fingers should be aligned again :D

- XCET44: pressure ****3.994**** bar, events ****x****, gas ****CO2****
- Run started: ****17-Aug-2026 18:35:59****
- Run finished: ****17-Aug-2026 20:22:00****
- Number of events: ****311189****
- DESY Table position: ****H: 35.344, V:1.340****

from here it doesn't seem like the beam is aligned with center.it (<http://center.it>) seems to be more hitting to the cal14. either cal14 is not parallel to the beam and the beam hits later on, or if it is parallel it is just too close. we will repeat the alignment, which is also necessary since the FS0 and FS1 is not as high as yesterday.



we were able to connect to the TimePix in the zone. we will try to put it into the beam just for couple of spills while trying to as much as possible not hitting to the katherine box.

We have aligned the DWC, XBPF and markers again and again!! After making more precise alignment we figured out that beam is not going through from the gap between CALs but passing through CAL14 and CAL18. So, our plan to make sure we have equal gap between both towers. Difference between distance of upstream of desy table to right tower and left tower >7 mm, needs to be corrected as well.

2026.08.16

- Run **1786864539** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL18**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50107**
 - **Initial look:**
 - Event rate from CESAR: approx **2832** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **132**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2759**, gas **CO2**
 - Run started: ****2026-08-16 09:15 ****
 - Run finished: ****2026-08-16 09:28 ****
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:41.755**
 - DESY table vertical position: **V:-3.388**
 - **Observation:** CAL18 IS CENTERED AROUND ~1000 QDC COUNT.

CAL18 IS DONE YAYY :)

we are doing CAL5 calibration again because the neighbour ECAL's are showing high counts in QDC

- Run **1786865850** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL5**

- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **59330**
- **Initial look:**
 - Event rate from CESAR: approx **2691** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **150**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2638**, gas **CO2**
 - Run started: ****2026-08-16 09:37 ****
 - Run finished: ****2026-08-16 09:52 ****
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:41.755**
 - DESY table vertical position: **V:-3.388**
- **Observation:** CAL5 IS CENTERED AROUND ~697 QDC COUNT.
the left right mean is -1.37+/-25.43mm
the up down mean is -0.81 +/-23.82mm

- Run **1786866862** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL5**

- beamfile: **001**
- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **55630**the version on disk (revert)?
- **Initial look:**
 - Event rate from CESAR: approx **4886** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill

- XCET44: pressure **0.308** bar, events **255**, gas **CO2**
- XCET48: pressure **1.005** bar, events **4717**, gas **CO2**
- Run started: **2026-08-16 09:54 **
- Run finished: **2026-08-16 10:05 **

○ **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **H:41.755**
- DESY table vertical position: **V:-3.388**

- **Observation:** CAL5 IS CENTERED AROUND ~1600 QDC COUNT.
the left right mean is -0.86 +/-21.94mm
the up down mean is -2.07 +/-20.26mm

● Run **1786867665** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL5**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **52678**

○ **Initial look:**

- Event rate from CESAR: approx **5229** events/spill
- Trigger rate from TDAQ: approx **1800** events/spill
- XCET44: pressure **0.308** bar, events **262**, gas **CO2**
- XCET48: pressure **1.005** bar, events **4876**, gas **CO2**
- Run started: **2026-08-16 10:07 **
- Run finished: **2026-08-16 10:18 **

○ **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **H:41.755**
- DESY table vertical position: **V:-3.388**

- **Observation:** CAL5 IS CENTERED AROUND ~3015 QDC COUNT.
the left right mean is -0.36 +/-20.49mm
the up down mean is -1.81 +/-19.14mm

CAL5 is done!!! :)

we are callibrating CAL4 because there was no FS0 and FS1 off earlier
we are going to repeat the CAL4 with FS0 and FS1 included to the DAQ

- Run **1786868952 : -3 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **51297**
 - **Initial look:**
 - Event rate from CESAR: approx **5502** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **232**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5111**, gas **CO2**
 - Run started: ****2026-08-16 10:29 ****
 - Run finished: ****2026-08-16 10:39 ****
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.470**
 - DESY table vertical position: **V:-3.696**
 - **Observation:** CAL4 IS CENTERED AROUND ~3119 QDC COUNT.
the left right mean is -0.35 +/-20.46mm
the up down mean is -1.84 +/-18.81mm
- Run **1786868952 : -2 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **53113**
 - **Initial look:**
 - Event rate from CESAR: approx **4897** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill

- XCET44: pressure **0.308** bar, events **258**, gas **CO2**
- XCET48: pressure **1.005** bar, events **4715**, gas **CO2**
- Run started: **2026-08-16 10:40 **
- Run finished: **2026-08-16 10:51 **
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.470**
 - DESY table vertical position: **V:-3.696**
- **Observation:** CAL4 IS CENTERED AROUND ~2107 QDC COUNT.
the left right mean is -0.92 +/-22.10mm
the up down mean is -1.49 +/-20.17mm
- Run **1786870373** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50544**
 - **Initial look:**
 - Event rate from CESAR: approx **5028** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **261**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4859**, gas **CO2**
 - Run started: **2026-08-16 10:52 **
 - Run finished: **2026-08-16 11:13 **
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.470**
 - DESY table vertical position: **V:-3.696**
 - **Observation:** CAL4 IS CENTERED AROUND ~1112 QDC COUNT.
the left right mean is -1.65 +/-25.39mm
the up down mean is -0.22 +/-24.02mm

CAL4 is done yayyyyyyy!! :) with the finger scintillators.

Changed DESY table position from horizontal 31.470 and vertical -3.696 to horizontal 30.468 and vertical -3.696 to test the cluster algorithm by making a controlled change.

- Run ****1786872460**** : **** -3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE****
- beamfile: ****002****
- beam momentum: **** -3** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****34870****
- ****Initial look****:
 - Event rate from CESAR: approx ****5414**** events/spill
 - Trigger rate from TDAQ: approx ****1800**** events/spill
 - XCET44: pressure ****0.308**** bar, events ****239****, gas ****CO2****
 - XCET48: pressure ****1.005**** bar, events ****5004****, gas ****CO2****
 - Run started: ****2026-08-16 11:27 ****
 - Run finished: ****2026-08-16 11:38 ****
- ****Calorimeters and DESY table****:
 - calorimeter gap: ****NO GAP****
 - DESY table horizontal position: ****H:30.468****
 - DESY table vertical position: ****V:-3.696****
- ****Observation****: CAL4 IS CENTERED AROUND ~3.1K QDC COUNT.
 the left right mean is -0.44 +/-20.43mm
 the up down mean is -1.33 +/-19.31mm

Changed DESY table position from horizontal 30.468 and vertical -3.696 to horizontal 29.467 and vertical -3.696 to test the cluster algorithm by making one more controlled change.

- Run **1786873403** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **35083**
 - **Initial look**:
 - Event rate from CESAR: approx **5450** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill

- XCET44: pressure **0.308** bar, events **246**, gas **CO2**
- XCET48: pressure **1.005** bar, events **5060**, gas **CO2**
- Run started: **2026-08-16 11:43 **
- Run finished: **2026-08-16 11:54 **
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:29.467**
 - DESY table vertical position: **V:-3.696**
- **Observation:** CAL4 IS CENTERED AROUND ~3150 QDC COUNT.
the left right mean is -0.73 +/-20.20mm
the up down mean is -2.01 +/-19.10mm

Changed DESY table position from horizontal 29.467 and vertical -3.696 to horizontal 28.465 and vertical -3.696 to test the cluster algorithm by making the third controlled change.

- Run **1786874231** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **143574**
- **Initial look:**
 - Event rate from CESAR: approx **5450** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **246**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5060**, gas **CO2**
 - Run started: **2026-08-16 11:57:11 **
 - Run finished: **2026-08-16 12:29:08 **
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:28.465**
 - DESY table vertical position: **V:-3.696**

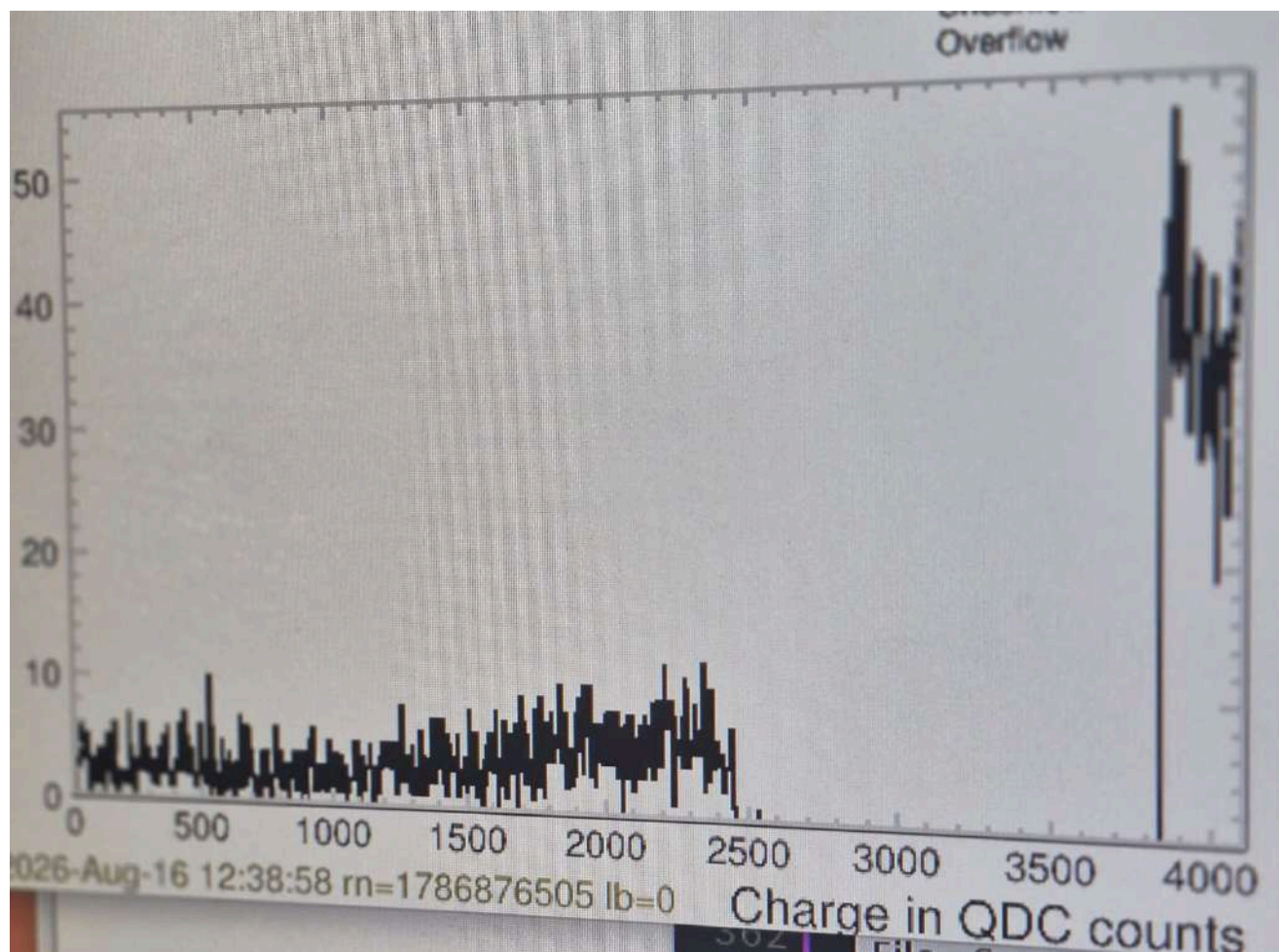
- **Observation:** CAL4 IS CENTERED AROUND ~3100 QDC COUNT.
the left right mean is -0.73 +/-20.20mm
the up down mean is -2.01 +/-19.10mm

Changed DESY table position from horizontal 28.465 and vertical -3.696 to horizontal 27.463 and vertical -3.696 to test the cluster algorithm by making the fourth controlled change.

- Run **1786876505** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **35288**
- **Initial look:**
 - Event rate from CESAR: approx **5450** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **246**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5060**, gas **CO2**
 - Run started: ****2026-08-16 12:35:05 ****
 - Run finished: ****2026-08-16 12:42:37 ****
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:27.463**
 - DESY table vertical position: **V:-3.696**
- **Observation:** CAL4 IS CENTERED AROUND ~3100 QDC COUNT.
the left right mean is -0.73 +/-20.20mm
the up down mean is -2.01 +/-19.10mm.

CAL1 IS AGAIN MAKING THAT AWKWARD SIGNAL AGAIN.



WE NEED TO CHANGE THE CAL1 TO CH17.

WE WENT INSIDE AND CHANGED CAL1 FROM CH1 TO CH17. LET'S REPEAT THE RUN.

- Run **1786877233** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **X**
 - **Initial look:**
 - Event rate from CESAR: approx **5450** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **246**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5060**, gas **CO2**
 - Run started: **2026-08-16 12:47:13**
 - Run finished: ****2026-08-16 X ****

- **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **H:27.463**
- DESY table vertical position: **V:-3.696**

during this run we've entered the zone to terminate ch0 and ch1 to understand what is going on with those channels.

- ****Observation****: CAL4 IS CENTERED AROUND ~3000 QDC COUNT.

the left right mean is -0.73 +/-20.20mm

the up down mean is -2.01 +/-19.10mm. Now the cal1-ch17 looks okay. but that means we



run number: 1786878223 total: 110754 finish 16-Aug-2026 13:27:56.

The ch0 and ch1 of QDC were reading unreasonably high values. So we discussed the



- **Run 1786882172 : -3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 1cm LOWER FROM THE CENTER OF CAL4**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **36781**
- **Initial look:**
 - Event rate from CESAR: approx **4755** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **219**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4433**, gas **CO2**
 - Run started: **2026-08-16 14:09:32**
 - Run finished: **2026-08-16 14:17:39**

- **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **H:31.476**
- DESY table vertical position: **V:-2.690**
- Run **1786883073** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 2cm LOWER FROM THE CENTER OF CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **35327**
 - **Initial look:**
 - Event rate from CESAR: approx **5383** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **225**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4564**, gas **CO2**
 - Run started: **2026-08-16 14:24:33**
 - Run finished: **2026-08-16 14:32:11**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:-1.694**
- Run **1786883648** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 3cm LOWER FROM THE CENTER OF CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **38658**
 - **Initial look:**
 - Event rate from CESAR: approx **5421** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill

- XCET44: pressure **0.308** bar, events **241**, gas **CO2**
- XCET48: pressure **1.005** bar, events **5005**, gas **CO2**
- Run started: **2026-08-16 14:34:08**
- Run finished: **2026-08-16 14:42:41**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:-0.696**
- **Run 1786884479 : -3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 4cm LOWER FROM THE CENTER OF CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **31766**
 - **Initial look:**
 - Event rate from CESAR: approx **5177** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **238**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4795**, gas **CO2**
 - Run started: **2026-08-16 14:47:59**
 - Run finished: **2026-08-16 14:54:48**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:0.310**
- **Run 1786885113 : -3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 5cm LOWER FROM THE CENTER OF CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**

- Trigger Condition: **S0S1**
- Number of events: **31766**
- **Initial look:**
 - Event rate from CESAR: approx **5389** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **258**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4981**, gas **CO2**
 - Run started: **2026-08-16 14:58:33**
 - Run finished: **2026-08-16 15:04**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:1.310**

After we started the high voltage, cal12 exceeded the set limit of current and tripped then we lowered it to 1500V but it tripped again. So we turned it off for a moment. Later due to the high current we changed the limit to 1475V and now it stopped tripping. We also changed the blanket with a black plastic.

- Run **1786887458** : **-3 GeV BEAM, TRYING TO TEST THE CLUSTERING CODE, TRYING TO UNDERSTAND HOW THE COUNTS OF NEIGHBORING CALORIMETERS CHANGE WHEN THE BEAM HITS 5cm LOWER FROM THE CENTER OF CAL4. WE'RE REDOING THE RUN BECAUSE PREVIOUS DATA WAS A BIT FAULTY.**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **35072**
 - **Initial look:**
 - Event rate from CESAR: approx **5320** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **232**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4931**, gas **CO2**
 - Run started: **2026-08-16 15:37:38**

- Run finished: **2026-08-16 15:45:51**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:1.310**
- Run **1786890116** : **-1 GeV BEAM, testing the clustering code at low energy.**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **32212**
 - **Initial look:**
 - Event rate from CESAR: approx **2826** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **147**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2490**, gas **CO2**
 - Run started: **2026-08-16 16:21:56**
 - Run finished: **2026-08-16 16:31:14**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:1.310**
- Run **1786890116** : **-1 GeV BEAM,testing the clustering code at low energy**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **33970**
 - **Initial look:**
 - Event rate from CESAR: approx **2277** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill

- XCET44: pressure **0.308** bar, events **147**, gas **CO2**
- XCET48: pressure **1.003** bar, events **2490**, gas **CO2**
- Run started: **2026-08-16 16:36:50**
- Run finished: **2026-08-16 16:46:51**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:0.310**
- Run **1786891837** : **-1 GeV BEAM, testing the clustering code at low energy**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **45998**
 - **Initial look:**
 - Event rate from CESAR: approx **2277** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **120**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **2733**, gas **CO2**
 - Run started: **16-08-2026 16:50:37**
 - Run finished: **16-08-2026 17:03:33**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:-0.689**
- Run **1786892766** : **-1 GeV BEAM, testing the clustering code at low energy**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **33117**
 - **Initial look:**

- Event rate from CESAR: approx **2736** events/spill
- Trigger rate from TDAQ: approx **1400** events/spill
- XCET44: pressure **0.308** bar, events **140**, gas **CO2**
- XCET48: pressure **1.003** bar, events **2673**, gas **CO2**
- Run started: **16-08-2026 17:06:06**
- Run finished: **16-08-2026 17:14:01**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:-1.689**
- Run **1786893316** : **-1 GeV BEAM, testing the clustering code at low energy**
- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **30446**
- **Initial look:**
 - Event rate from CESAR: approx **2736** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **150**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **2746**, gas **CO2**
 - Run started: **16-Aug-2026 17:15:16**
 - Run finished: **16-Aug-2026 17:22:36**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:31.476**
 - DESY table vertical position: **V:-2.689**
- Run **1786894793** : **-1 GeV BEAM, testing the clustering code at low energy**
- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**

- Number of events: **49257**
- **Initial look:**
 - Event rate from CESAR: approx **2655** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **300**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **5000**, gas **CO2**
 - Run started: **16-Aug-2026 17:39:53**
 - Run finished: **16-Aug-2026 17:51:50**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:30.473**
 - DESY table vertical position: **V:-3.696**
- Run **1786895892** : **-1 GeV BEAM, testing the clustering code at low energy**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **367090**
 - **Initial look:**
 - Event rate from CESAR: approx **2655** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **300**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **5000**, gas **CO2**
 - Run started: **16-Aug-2026 17:58**
 - Run finished: **16-Aug-2026 19:43**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:29.475**
 - DESY table vertical position: **V:-3.696**
- Run **1786903064** : **-1 GeV BEAM,**
 - beamfile: **000**

- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **36740**
- **Initial look:**
 - Event rate from CESAR: approx **2655** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **300**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **5000**, gas **CO2**
 - Run started: **16-Aug-2026 19:57**
 - Run finished: **16-Aug-2026 20:09**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:28.479**
 - DESY table vertical position: **V:-3.696**
- Run **1786903854** : **-1 GeV BEAM, testing the clustering code at low energy**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **x**
 - **Initial look:**
 - Event rate from CESAR: approx **2655** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **300**, gas **CO2**
 - XCET48: pressure **1.003** bar, events **5000**, gas **CO2**
 - Run started: **16-Aug-2026 20:10**
 - Run finished: **16-Aug-2026 20:27**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:27.476**
 - DESY table vertical position: **V:-3.696**

when this run is finished, it means we are done with all cluster calibration runs.

we've got curious about the beam spot movements and beam spot center at high and low energies. we have pos hadron +15 GeV, +10 GeV, + 5 GeV +1 GeV Beam files. With each one of them we will check how much the beam spot moves.

Beam Energy	Plane	Mean Position [mm]	RMS Width [mm]	Cut-off
+15 GeV	SJ	-0.3	3.9	-
+15 GeV	UD	-0.7	5.2	-
+10 GeV	SJ	-5.7	5.4	57%
+10 GeV	UD	+6.2	5.4	60%
+5 GeV	SJ	+3.8	5.3	77%
+5 GeV	UD	+3.5	5.7	73%
+3 GeV	SJ	-4.2	10.2	58%
+3 GeV	UD	+0.6	10.6	53%
+3 GeV (second measurement)	SJ	-4.1	5.1	85%
+3 GeV (second measurement)	UD	+2.1	4.9	85%
+1 GeV	SJ	-	-	No clear beam center
+1 GeV	UD	-	-	No clear beam center

► Details

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- Run ****1786905364**** : ****15 GeV BEAM**, TO SEE HOW THE WIDTH OF BEAM CHANGES WITH ENERGY
- beamfile: ****003****
- beam momentum: ****15** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****14699****

- Run ****1786906157**** : ****10 GeV BEAM**, TO SEE HOW THE WIDTH OF BEAM CHANGES WITH ENERGY
- beamfile: ****004****
- beam momentum: ****10** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****52821****

- Run ****1786906692**** : ****5 GeV BEAM**, TO SEE HOW THE WIDTH OF BEAM CHANGES WITH ENERGY
- beamfile: ****007****
- beam momentum: ****5** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****37615****

- Run ****1786907349**** : ****1 GeV BEAM**, TO SEE HOW THE WIDTH OF BEAM CHANGES WITH ENERGY
- beamfile: ****010****
- beam momentum: ****1** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****592****

- Run ****1786907647**** : ****3 GeV BEAM**, TO SEE HOW THE WIDTH OF BEAM CHANGES WITH ENERGY
- beamfile: ****008****
- beam momentum: ****3** GeV**
- Trigger Condition: ****S0S1****
- Number of events: ****70k****

we will take our first physics run yihhuu. now setting our cherenkov threshold values. one of them will be set XCET48 to 2 bar to tag e,mu. and XCET48 to 4 bar to tag e mu pi. we have also increased the voltage on the cal12 to 1525 also not to repeat the calibration. But we have changed the cover with something lighter, some thing black, and something smells terrible. but it is designed for dark room experiments, so we believe it is still very helpful.

- Run **1786914869** : **+3 GeV BEAM**

- beamfile: **008**
- beam momentum: **+3 GeV**
- Trigger Condition: **S0S1FS0FS1**
- Number of events: **2011309**
- **Initial look:**
 - Event rate from CESAR: approx x^* events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **x**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **x**, gas **CO2**
 - Run started: **2026-08-16 23:14**
 - Run finished: **2026-08-17 10:07**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **36.151**
 - DESY table vertical position: **1.180**
- **Observation:** we put 3 cm Graphite target. Distance between US of Target to US of CALs=50 cm!

we go home! we must sleeppppppppppppppppp

2026.08.15

- Run **1786744833** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL13**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **57401**
 - **Initial look:**
 - Event rate from CESAR: approx $5K^*$ events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill

- XCET44: pressure **0.308** bar, events **311**, gas **CO2**
- XCET48: pressure **1.005** bar, events **5034**, gas **CO2**
- Run started: **2026-08-14 00:00:33**
- Run finished: **2026-08-14 00:12:50**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **51.414**
 - DESY table vertical position: **15.768**
- **Observation:** CAL13 IS CENTERED AROUND **2400** QDC COUNT.

CAL13 is finished

- Run **1786745756** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL11**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **52605**
 - **Initial look:**
 - Event rate from CESAR: approx **5K*** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **311**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5034**, gas **CO2**
 - Run started: **2026-08-15 00:15:56**
 - Run finished: **2026-08-15 00:27:00**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **51.394**
 - DESY table vertical position: **6.271**
 - **Observation:** CAL11 IS CENTERED AROUND **2600** QDC COUNT.
- Run **1786746501** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL11**
 - beamfile: **001**

- beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50280**
 - **Initial look:**
 - Event rate from CESAR: approx **5K*** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **311**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5034**, gas **CO2**
 - Run started: **2026-08-15 00:28:21**
 - Run finished: **2026-08-15 00:39:45**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **51.394**
 - DESY table vertical position: **6.271**
 - **Observation:** CAL11 IS CENTERED AROUND **1800** QDC COUNT.
- Run **1786747305** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL11**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **54196**
 - **Initial look:**
 - Event rate from CESAR: approx **3K*** events/spill
 - Trigger rate from TDAQ: approx **1.3K** events/spill
 - XCET44: pressure **0.308** bar, events **197**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **3.9k**, gas **CO2**
 - Run started: **2026-08-15 00:41:45**
 - Run finished: **2026-08-15 00:57:00**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **51.394**

- DESY table vertical position: **6.271**

- **Observation:** CAL11 IS CENTERED AROUND **900** QDC COUNT.

CAL11 is finished

- Run **1786748346** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL14**

- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **49937**
- **Initial look:**
 - Event rate from CESAR: approx **3K*** events/spill
 - Trigger rate from TDAQ: approx **1.3K** events/spill
 - XCET44: pressure **0.308** bar, events **197**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **3.9k**, gas **CO2**
 - Run started: **2026-08-15 00:59:06**
 - Run finished: ****2026-08-15 01:13:18**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.418**
 - DESY table vertical position: **6.398**
- **Observation:** CAL14 IS CENTERED AROUND **1200** QDC COUNT.

- Run **1786750147** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL14**

- beamfile: **001**
- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50381**
- **Initial look:**
 - Event rate from CESAR: approx **5.3K*** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill

- XCET44: pressure **0.308** bar, events **197**, gas **CO2**
- XCET48: pressure **1.005** bar, events **5.1k**, gas **CO2**
- Run started: **2026-08-15 01:29:07**
- Run finished: **2026-08-15 01:40:14**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.418**
 - DESY table vertical position: **6.398**
- **Observation:** CAL14 IS CENTERED AROUND **2300** QDC COUNT.
- Run **1786750887** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL14**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **2.6m**
 - **Initial look:**
 - Event rate from CESAR: approx **4.9K*** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **197**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4.6k**, gas **CO2**
 - Run started: **2026-08-15 01:41:27**
 - Run finished: **2026-08-15 09:35**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.418**
 - DESY table vertical position: **6.398**
 - **Observation:** CAL14 IS CENTERED AROUND **3300** QDC COUNT.

unfort no beam:(

CAL14 finished

- Run **1786779710** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL2**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **57950**
 - **Initial look:**
 - Event rate from CESAR: approx 5517* events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **238**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4973**, gas **CO2**
 - Run started: **2026-08-15 09:41**
 - Run finished: **2026-08-15 09:54**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.197**
 - DESY table vertical position: **6.333**
 - **Observation:** CAL2 IS CENTERED AROUND **3200** QDC COUNT.
- Run **1786780617** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL2**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **x54909**
 - **Initial look:**
 - Event rate from CESAR: approx 5218* events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **266**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5042**, gas **CO2**
 - Run started: **2026-08-15 09:42**
 - Run finished: **2026-08-15 10:09**
 - **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **31.197**
- DESY table vertical position: **6.333**
- **Observation:** CAL2 IS CENTERED AROUND **2200** QDC COUNT.
- Run **1786781408** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL2**
- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **55253**
- **Initial look:**
 - Event rate from CESAR: approx **5200** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **162**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **3005**, gas **CO2**
 - Run started: **2026-08-15 10:10**
 - Run finished: **2026-08-15 10:25**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.197**
 - DESY table vertical position: **6.333**
- **Observation:** CAL2 IS CENTERED AROUND **1200** QDC COUNT.

CAL2 has finished

- Run **1786782465** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL0**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50390**
 - **Initial look:**

- Event rate from CESAR: approx **5200** events/spill
- Trigger rate from TDAQ: approx **1.8k** events/spill
- XCET44: pressure **0.308** bar, events **162**, gas **CO2**
- XCET48: pressure **1.005** bar, events **3005**, gas **CO2**
- Run started: **2026-08-15 10:27**
- Run finished: **2026-08-15 10:42**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.059**
 - DESY table vertical position: **6.421**
- **Observation:** CALO IS CENTERED AROUND **1200** QDC COUNT.
- Run **1786787514** : **-2 GeV BEAM, TRYING TO CALIBRATE CALO**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **52627**
 - **Initial look:**
 - Event rate from CESAR: approx **5000** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **162**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4.8k**, gas **CO2**
 - Run started: **2026-08-15 11:51:54**
 - Run finished: **2026-08-15 12:06:20**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.059**
 - DESY table vertical position: **6.421**
 - **Observation:** CALO IS CENTERED AROUND **2000** QDC COUNT.
- Run **1786788903** : **-3 GeV BEAM, TRYING TO CALIBRATE CALO**
 - beamfile: **002**

- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **75301**
- **Initial look:**
 - Event rate from CESAR: approx **5,1k** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **246**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4.7k**, gas **CO2**
 - Run started: **2026-08-15 12:15**
 - Run finished: ****2026-08-15 12:35 ****
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.059**
 - DESY table vertical position: **6.421**
- **Observation:** CALO IS CENTERED AROUND **3000** QDC COUNT.

CALO has finished

- Run **1786790499** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL10**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **68453**
 - **Initial look:**
 - Event rate from CESAR: approx **5,1k** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **246**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4.7k**, gas **CO2**
 - Run started: **2026-08-15 12:41**
 - Run finished: ****2026-08-15 13:03:34 ****
 - **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **21.683**
- DESY table vertical position: **-13.166**
- **Observation:** CAL10 IS CENTERED AROUND **2400** QDC COUNT.
- Run **1786791937** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL10**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **56256**
 - **Initial look:**
 - Event rate from CESAR: approx **5211k** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **230**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5019**, gas **CO2**
 - Run started: **2026-08-15 13:05**
 - Run finished: ****2026-08-15 13:26 ****
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.683**
 - DESY table vertical position: **-13.166**
 - **Observation:** CAL10 IS CENTERED AROUND **1700** QDC COUNT.
- Run **1786793439** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL10**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: ********
 - **Initial look:**
 - Event rate from CESAR: approx **2.7k** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill

- XCET44: pressure **0.308** bar, events **134**, gas **CO2**
- XCET48: pressure **1.005** bar, events **2708**, gas **CO2**
- Run started: **2026-08-15 13:30**
- Run finished: ****2026-08-15 13:49 ****
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.683**
 - DESY table vertical position: **-13.166**
- **Observation:** CAL10 IS CENTERED AROUND **900** QDC COUNT.

CAL10 has finishedd!!

- Run **1786794828** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL5**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **53044**
 - **Initial look:**
 - Event rate from CESAR: approx **2.9k** events/spill
 - Trigger rate from TDAQ: approx **1.2k** events/spill
 - XCET44: pressure **0.308** bar, events **139**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2867**, gas **CO2**
 - Run started: **2026-08-15 13:53**
 - Run finished: ****2026-08-15 14:12 ****
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.856**
 - DESY table vertical position: **-12.986**
 - **Observation:** CAL5 IS CENTERED AROUND **1090** QDC COUNT.
- Run **1786796090** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL5**
 - beamfile: **001**

- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **51109**
- **Initial look:**
 - Event rate from CESAR: approx **5k** events/spill
 - Trigger rate from TDAQ: approx **1.6k** events/spill
 - XCET44: pressure **0.308** bar, events **219**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4837**, gas **CO2**
 - Run started: ****2026-08-15 14:14 ****
 - Run finished: ****2026-08-15 14:28 ****
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.856**
 - DESY table vertical position: **-12.986**
- **Observation:** CAL5 IS CENTERED AROUND **2070** QDC COUNT.
 - Run **1786797061** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL5**
- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **52630**
- **Initial look:**
 - Event rate from CESAR: approx **5.4k** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **233**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5018**, gas **CO2**
 - Run started: **2026-08-15 14:31**
 - Run finished: **2026-08-15 14:44**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.856**

- DESY table vertical position: **-12.986**

- **Observation:** CAL5 IS CENTERED AROUND **3089** QDC COUNT.

CAL5 has finished YAY!!

- Run **1786798201** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL1**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **51149**
- **Initial look:**
 - Event rate from CESAR: approx **5.5K** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **288**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5197**, gas **CO2**
 - Run started: **2026-08-15 14:50**
 - Run finished: **2026-08-15 15:03**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **42.114**
 - DESY table vertical position: **-12.986**
- **Observation:** CAL1 IS CENTERED AROUND **2308** QDC COUNT.

- Run **1786799056** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL1**

- beamfile: **001**
- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50791**
- **Initial look:**
 - Event rate from CESAR: approx **5.2K** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill

- XCET44: pressure **0.308** bar, events **227**, gas **CO2**
- XCET48: pressure **1.005** bar, events **5120**, gas **CO2**
- Run started: **2026-08-15 15:04**
- Run finished: **2026-08-15 15:18**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **42.114**
 - DESY table vertical position: **-12.986**
- **Observation:** CAL1 IS CENTERED AROUND **1516** QDC COUNT.
- Run **1786800039** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL1**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50235**
 - **Initial look:**
 - Event rate from CESAR: approx **2.8K** events/spill
 - Trigger rate from TDAQ: approx **1.4K** events/spill
 - XCET44: pressure **0.308** bar, events **125**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2703**, gas **CO2**
 - Run started: **2026-08-15 15:20**
 - Run finished: **2026-08-15 15:38**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **42.114**
 - DESY table vertical position: **-12.986**
 - **Observation:** CAL1 IS CENTERED AROUND **820** QDC COUNT.

CAL1 FINISHED

- Run **1786801341** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL12**
 - beamfile: **000**

- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **54129**
- **Initial look:**
 - Event rate from CESAR: approx **2748** events/spill
 - Trigger rate from TDAQ: approx **1.4K** events/spill
 - XCET44: pressure **0.308** bar, events **131**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2789**, gas **CO2**
 - Run started: **2026-08-15 15:42**
 - Run finished: **2026-08-15 16:01:43**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **42.114**
 - DESY table vertical position: **-12.986**
- **Observation:** CAL12 IS CENTERED AROUND **750** QDC COUNT.
- Run **1786802590** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL12**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50889**
 - **Initial look:**
 - Event rate from CESAR: approx **4783** events/spill
 - Trigger rate from TDAQ: approx **1.7K** events/spill
 - XCET44: pressure **0.308** bar, events **201**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4627**, gas **CO2**
 - Run started: **2026-08-15 16:03:10**
 - Run finished: **2026-08-15 16:17:43**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **42.114**

- DESY table vertical position: **-12.986**

- **Observation:** CAL12 IS CENTERED AROUND **1400** QDC COUNT.

We had ~2k counts with CAL12 in June Testbeam. Now it is less. This CAL12 also get tripped once today. It looks fine now, but let's follow it up.

- Run **1786803907** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL12**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50661**
- **Initial look:**
 - Event rate from CESAR: approx **4989** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **200**, gas **CO2**
 - XCET48: pressure **1.004** bar, events **4642**, gas **CO2**
 - Run started: **2026-08-15 16:25:07**
 - Run finished: **2026-08-15 16:38:51**

- **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **42.114**
- DESY table vertical position: **-12.986**

- **Observation:** CAL12 IS CENTERED AROUND **2000** QDC COUNT.
The QDC count is lower than the June test beam.

Cal12 has finished

- Run **1786805198** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL17**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**

- Number of events: **52349**
- **Initial look:**
 - Event rate from CESAR: approx **5042** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **239**, gas **CO2**
 - XCET48: pressure **1.004** bar, events **4666**, gas **CO2**
 - Run started: **2026-08-15 16:46:38**
 - Run finished: **2026-08-15 17:00:25**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.695**
 - DESY table vertical position: **-3.697**
- **Observation:** CAL17 IS CENTERED AROUND **2700** QDC COUNT.
The QDC count is lower than the June test beam.
- Run **1786806354** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL17**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **51081**
 - **Initial look:**
 - Event rate from CESAR: approx **5015** events/spill
 - Trigger rate from TDAQ: approx **1.8k** events/spill
 - XCET44: pressure **0.308** bar, events **236**, gas **CO2**
 - XCET48: pressure **1.004** bar, events **4866**, gas **CO2**
 - Run started: **2026-08-15 17:05:54**
 - Run finished: **2026-08-15 17:23:05**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.695**
 - DESY table vertical position: **-3.697**

- **Observation:** CAL17 IS CENTERED AROUND **1900** QDC COUNT.
The QDC count is lower than the June test beam.
- Run **1786807867** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL17**
- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **58556**
- **Initial look:**
 - Event rate from CESAR: approx **2775** events/spill
 - Trigger rate from TDAQ: approx **1.4k** events/spill
 - XCET44: pressure **0.308** bar, events **128**, gas **CO2**
 - XCET48: pressure **1.004** bar, events **2742**, gas **CO2**
 - Run started: **2026-08-15 17:31:07**
 - Run finished: **2026-08-15 17:52**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.695**
 - DESY table vertical position: **-3.697**
- **Observation:** CAL17 IS CENTERED AROUND **1000** QDC COUNT.
The QDC count is slightly lower than the June test beam.

18:33 we have brought these signals (CAL12 & CAL17) into the control through the patch panel to see if there is something wrong with the timing of the signal. It seems they are both in the gate perfectly well. Now there is one remaining possibility with it which is that they are not aligned well. We And also very interestingly when we are aiming for cal12, cal0 is reading a lot of values. We've decided its time for us to give a break, because this is not making any sense to us.

We need the entire monitors to begin with.

And we have checked the cal1 an cal7 and cal18, with the 3 GeV beam, cal7 seems to read quite high value. we will do the alignment of the cal12 again. It didn't seem way too amazing, and we decided to center the circle. The new H: 52.084, V:-13.246. After this the cal7 that is under the cal12 was still reading a lot of values. We will just lower the DESY 1 cm to see the effect. And cal0 is completely crazy at this point. You can see the data at: **1786820108**.

we are at H: 52.084, V:-14.245 we take it! We've recorded about 30k particles, and it seems a bit higher, however cal7 is still counting a considerable amount. We will lower the desy table another cm, you can check this data at **1786820649**.

now we are at H:52.084, V:-15.247. Let's start another run. You can see the run at **1786821153**. It seems the lower and the left detectors are still counting. OK the left cal1 is counting, but why the cal7 is still counting!!! I'll go 1 more cm lower. cal0 is still completely crazy.

one at H:52.084. V:-16.248. Starting another run to see the monitors. It didn't matter much, it can be seen at **1786821668**. We will increase the HV.

We've made the HV 1525 V, don't tell it to anybody. It can be seen **1786822004**. It got a bit higher but

We wanted to check the neighbor values on the previous calibration (june) procedure. it seems the cal12 which is the problematic lower counting cal at that point had well centered beam. what we will do now is that we will convert all of our files (august) in to ROOT format (with old DWC calibration values, this should be checked again later on), so that we make sure on our calibration beam is centered enough.

meanwhile we will do the calibration of the DWCs.

This signal is our fake signal for the calibration of the DWCs.

1 kHz

Pulse 100 mVpp, 0 mV offset

120 ns width, ramp up 10 ns, ramp down 60 ns.

We've noticed that DWC0 is incredibly noisy. Firstly it is better to do the calibration without HV! secondly 2700 is either too high or the threshold of the discriminator is too low. At 2600 the detector calms down.

Trigger is BUSY & AH0&AV0.

DWC0 LU: 1786828413

DWC0 C: 1786828126

DWC0 RD: 1786828280

Trigger is BUSY & AH1&AV1.

DWC1 LU: 1786828856

DWC1 C: 1786828723

DWC1 RD: 1786828590

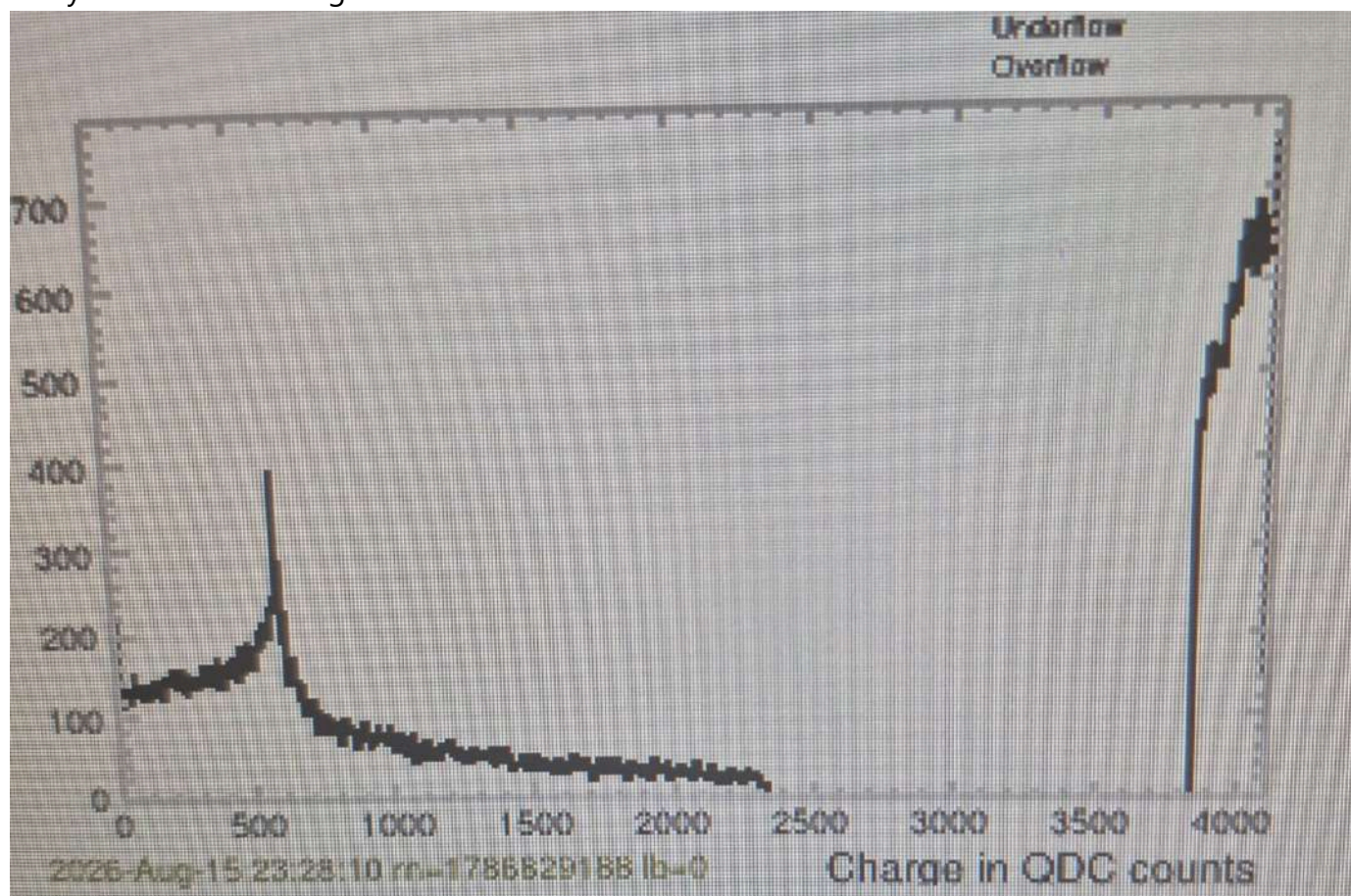
Pedastal runs:

HV ON, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : 1786828994

HV OFF, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : 1786829188

switching to our normal trigger configuration, S0S1.

Okay Cal0 is measuring this without the HV!



this is soo unacceptable. I'll try to check some cables.

I've noticed that the cable that goes into the delay unit is a bit loose. but I'd be shocked if it is because of that one. let's turn the beam on, and HV will stay OFF to see if it is still crayay. The run can be seen at **1786829938** cal0 is still completely crazy. On the oscilloscope the the signal that goes into the delay unit didn't seem to cause a problem. let's turn off the beam and see the signal that comes out of the delay unit, otherwise it is the QDC cable that is the problematic one! The run number for this trial is 1786829938. Now we've pushed the QDC adaptor, it didn't feel like it was loose, however let's see the cal0. as disgusting as before. :(run number can be seen at 1786830571. We need to get diabolical and get the other adaptor and measure on channel17.

****\overline{Pedastal runs with the new Ch16 which is where the cal0 is at from now on! this is actually not important that cal0 is at ch16. because ch16 is already also taken, on the previous runs, so they should be taken into care, not these. **:**
 HV ON, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : x
 HV OFF, Trigger BUSY & AH1&AV1, Fake signal on DWC1 : 1786832059}

now cal0 ch16 seems okay. but we need to repeat the calibration runs for it, we need to move the desy table in that direction. now the H:21.061, V:6.423.

- Run **1786833136** : **-3 GeV BEAM, TRYING TO CALIBRATE CALO**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **55074**
 - **Initial look:**
 - Event rate from CESAR: approx **5373** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **313**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4972**, gas **CO2**
 - Run started: ****2026-08-16 00:32:16 ****
 - Run finished: **2026-08-16 00:49:02**
 - **Calorimeters and DESY table:** H:21.094 V:16.233
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:21.061**
 - DESY table vertical position: **6.423**
 - **Observation:** CALO IS CENTERED AROUND ~2800 QDC COUNT. Seems a bit less but nothing todo.
- Run **1786834513** : **-2 GeV BEAM, TRYING TO CALIBRATE CALO**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **30588**
 - **Initial look:**
 - Event rate from CESAR: approx **5021** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **342**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4850**, gas **CO2**
 - Run started: ****2026-08-16 00:55:13 ****
 - Run finished: **2026-08-16 01:03:30**

- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:21.061**
 - DESY table vertical position: **6.423**
- **Observation:** CAL0 IS CENTERED AROUND ~1900 QDC COUNT. Seems a bit less but nothing todo.
- Run **1786835122** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL0**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **48433**
 - **Initial look:**
 - Event rate from CESAR: approx **2921** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **207**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2872**, gas **CO2**
 - Run started: ****2026-08-16 01:05:22 ****
 - Run finished: **2026-08-16 01:22:36**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:21.061**
 - DESY table vertical position: **6.423**
 - **Observation:** CAL0 IS CENTERED AROUND ~1000 QDC COUNT.

cal0 is finished again. moving on the cal7, changing the desy table position.

- Run **1786836376** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL7**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**

- Number of events: **41229**
- **Initial look:**
 - Event rate from CESAR: approx **2921** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **207**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2872**, gas **CO2**
 - Run started: ****2026-08-16 01:05:22 ****
 - Run finished: **2026-08-16 01:41:22**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:51.773**
 - DESY table vertical position: **-3.436**
- **Observation:** CAL7 IS CENTERED AROUND ~1100 QDC COUNT.
- Run **1786837396** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL7**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **50k**
 - **Initial look:**
 - Event rate from CESAR: approx **5021** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **342**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4850**, gas **CO2**
 - Run started: ****2026-08-16 00:55:13 ****
 - Run finished: **2026-08-16 01:57:08**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:51.773**
 - DESY table vertical position: **-3.436**
 - **Observation:** CAL7 IS CENTERED AROUND ~2100 QDC COUNT.

- Run **1786838309** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL7**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **56611**
- **Initial look:**
 - Event rate from CESAR: approx **5373** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **313**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4972**, gas **CO2**
 - Run started: ****2026-08-16 01:58:29 ****
 - Run finished: **2026-08-16 02:13:47**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:51.773**
 - DESY table vertical position: **-3.436**
- **Observation:** CAL7 IS CENTERED AROUND ~3000 QDC COUNT.

cal7 is also finished, moving onto cal18. moved the desy table to H:41.755, V:-3.388.

- Run **1786839384** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL18**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50965**
- **Initial look:**
 - Event rate from CESAR: approx **5373** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **313**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4972**, gas **CO2**
 - Run started: ****2026-08-16 02:16:24 ****

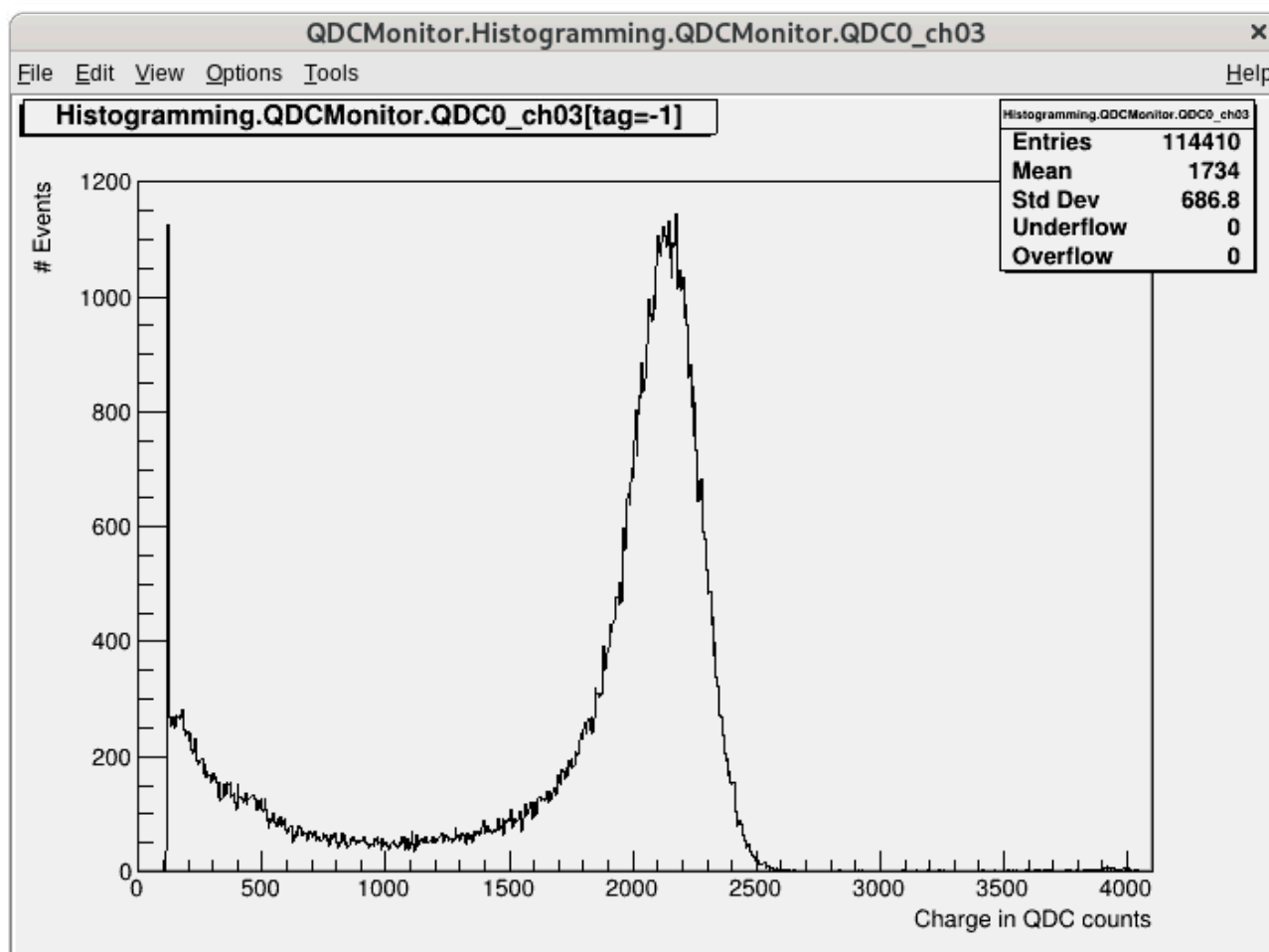
- Run finished: **2026-08-16 02:30:01**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:41.755**
 - DESY table vertical position: **V:-3.388**
- **Observation:** CAL18 IS CENTERED AROUND ~2800 QDC COUNT.
- Run **1786840283** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL18**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **1.8m**
 - **Initial look:**
 - Event rate from CESAR: approx **5073** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **377**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **4902**, gas **CO2**
 - Run started: ****2026-08-16 02:31:23 ****
 - Run finished: **2026-08-16 09:12**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **H:41.755**
 - DESY table vertical position: **V:-3.388**
 - **Observation:** CAL18 IS CENTERED AROUND ~2000 QDC COUNT.

2026.08.14

Continuation from last night...

- Run **1786662364** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **001**
 - beam momentum: **-2 GeV**

- Trigger Condition: **S0S1**
- Number of events: **1.86m**
- **Initial look:**
 - Event rate from CESAR: approx **** events/spill
 - Trigger rate from TDAQ: approx **1700** events/spill
 - XCET44: pressure ?? bar, events ?, gas **CO2**
 - XCET48: pressure **1** bar, events **5159**, gas **CO2**
 - Run started: **2026-08-14 01:06:04**
 - Run finished: **2026-08-14 09:29:13**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.472**
 - DESY table vertical position: **-3.696**
- **Observation:** XCET44 IS MISSING!!! CAL4 IS CENTERED AROUND 1734 QDC COUNT. Here is the SS below to see QDC counts from CAL4:



CALIBRATION RUNS:

- Run **1786708920** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL8**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **55074**
 - **Initial look:**
 - Event rate from CESAR: approx **5167** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **372**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **7235**, gas **CO2**
 - Run started: ****2026-08-14 14:02 ****
 - Run finished: **2026-08-14 14:33**
 - **Calorimeters and DESY table:** H:21.094 V:16.233
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.094**
 - DESY table vertical position: **16.233**
 - **Observation:** XCET44 came back! CAL8 IS CENTERED AROUND ~2000 QDC COUNT.

Warning: This run is a bit suspicious, we are not fully sure if CESAR has upload the - 3 GeV beam file correctly or if it stuck in 2 GeV run.

- Run **1786711761**: **-3 GeV BEAM, TRYING TO CALIBRATE CAL8**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **51008**
 - **Initial look:**
 - Event rate from CESAR: approx **5355** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure **0.308** bar, events **376**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **7566**, gas **CO2**

- Run started: **2026-08-14 14:49**
- Run finished: **2026-08-14 15:08**
- **Calorimeters and DESY table:** H:21.094 V:16.233
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.094**
 - DESY table vertical position: **16.233**
- **Observation:** CAL8 IS CENTERED AROUND ~2600 QDC COUNT.
- Run **1786713224** : **-2 GeV** BEAM, TRYING TO CALIBRATE CAL8**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **51913**
 - **Initial look:**
 - Event rate from CESAR: approx ~**5k** events/spill
 - Trigger rate from TDAQ: approx **1600** events/spill
 - XCET44: pressure **0.308** bar, events **349**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **7210**, gas **CO2**
 - Run started: **2026-08-14 15:13**
 - Run finished: **2026-08-14 15:32**
 - **Calorimeters and DESY table:** H:21.094 V:16.233
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.094**
 - DESY table vertical position: **16.233**
 - **Observation:** CAL8 IS CENTERED AROUND 1681 QDC COUNT.
- Run **1786714515** : **-1 GeV** BEAM, TRYING TO CALIBRATE CAL8
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **54535**

- **Initial look:**
 - Event rate from CESAR: approx **2835** events/spill
 - Trigger rate from TDAQ: approx **1200** events/spill
 - XCET44: pressure **0.308** bar, events **128**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **2786**, gas **CO2**
 - Run started: **2026-08-14 15:35**
 - Run finished: **2026-08-14 15:57**
- **Calorimeters and DESY table:** H:21.094 V:16.233
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **21.094**
 - DESY table vertical position: **16.233**
- **Observation:** CAL8 IS CENTERED AROUND **** QDC COUNT.

CAL8 is finished

- Run **1786723178** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL19**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **46714**
 - **Initial look:**
 - Event rate from CESAR: approx **2846** events/spill
 - Trigger rate from TDAQ: approx **1400** events/spill
 - XCET44: pressure **0.308** bar, events **289**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5507**, gas **CO2**
 - Run started: **2026-08-14 17:59**
 - Run finished: **2026-08-14 18:23:09**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.501**
 - DESY table vertical position: **15.943**

- **Observation:** CAL19 IS CENTERED AROUND **800** QDC COUNT.
- Run **1786724681** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL19**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **49674**
 - **Initial look:**
 - Event rate from CESAR: approx *5011** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **450**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5243**, gas **CO2**
 - Run started: **2026-08-14 18:24**
 - Run finished: **2026-08-14 18:36:15*
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.501**
 - DESY table vertical position: **15.943**
 - **Observation:** CAL19 IS CENTERED AROUND **1200** QDC COUNT.
- Run **1786725412** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL19**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **47546**
 - **Initial look:**
 - Event rate from CESAR: approx *5841** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **327**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5409**, gas **CO2**
 - Run started: **2026-08-14 18:36**
 - Run finished: **2026-08-14 18:47*

- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.501**
 - DESY table vertical position: **15.943**
- **Observation:** CAL19 IS CENTERED AROUND **1700** QDC COUNT.

CAL19 is finished

- Run **1786726168** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL9**

- beamfile: **002**
- beam momentum: **-3 GeV**
- Trigger Condition: **S0S1**
- Number of events: **~600000**
- **Initial look:**
 - Event rate from CESAR: approx 5841* events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **327**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5409**, gas **CO2**
 - Run started: **2026-08-14 18:49**
 - Run finished: **2026-08-14 22:34:57*

- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.551**
 - DESY table vertical position: **15.943**
- **Observation:** CAL9 IS CENTERED AROUND **3500** QDC COUNT.

- Run **1786739830** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL9**

- beamfile: **001**
- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **92689**

- **Initial look:**
 - Event rate from CESAR: approx 5211* events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **311**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **5034**, gas **CO2**
 - Run started: **2026-08-14 22:37:10**
 - Run finished: **2026-08-14 23:10:49**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.551**
 - DESY table vertical position: **15.943**
- **Observation:** CAL9 IS CENTERED AROUND **2300** QDC COUNT.
- Run **1786741940** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL9**
 - beamfile: **000**
 - beam momentum: **-1 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **51918**
 - **Initial look:**
 - Event rate from CESAR: approx ~3k events/spill
 - Trigger rate from TDAQ: approx **1.3k** events/spill
 - XCET44: pressure **0.308** bar, events ~280, gas **CO2**
 - XCET48: pressure **1.005** bar, events ~2890, gas **CO2**
 - Run started: **2026-08-14 23:12:20**
 - Run finished: **2026-08-14 23:28:49**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **41.551**
 - DESY table vertical position: **15.943**
 - **Observation:** CAL9 IS CENTERED AROUND **1200** QDC COUNT.

CAL9 is finished

- Run **1786743090** : **-1 GeV BEAM, TRYING TO CALIBRATE CAL13**

- beamfile: **000**
- beam momentum: **-1 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50587**
- **Initial look:**
 - Event rate from CESAR: approx **~3k** events/spill
 - Trigger rate from TDAQ: approx **1.3k** events/spill
 - XCET44: pressure **0.308** bar, events **~300**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **~3k**, gas **CO2**
 - Run started: **2026-08-14 23:31:30**
 - Run finished: **2026-08-14 23:45:25**
- **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **51.414**
 - DESY table vertical position: **15.768**
- **Observation:** CAL13 IS CENTERED AROUND **900** QDC COUNT.

- Run **1786744038** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL13**

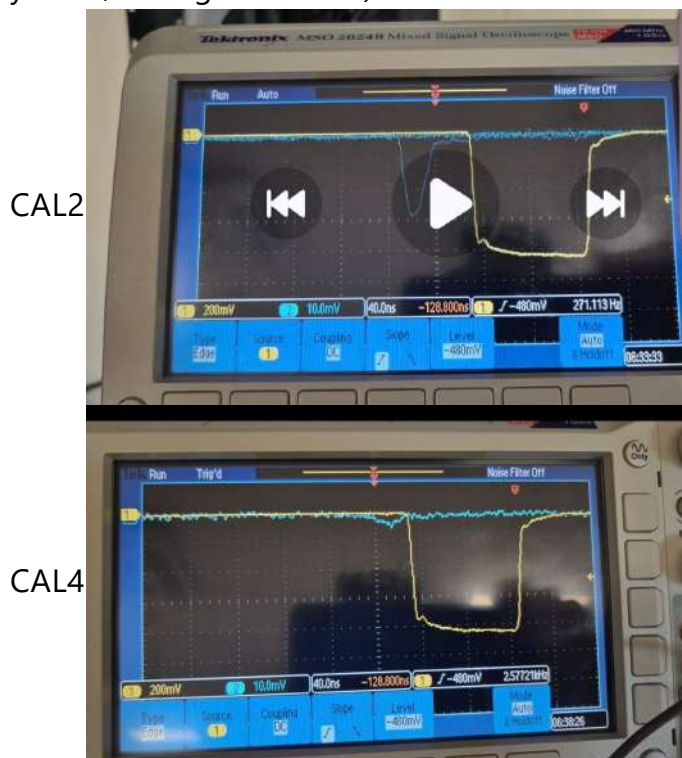
- beamfile: **001**
- beam momentum: **-2 GeV**
- Trigger Condition: **S0S1**
- Number of events: **50373**
- **Initial look:**
 - Event rate from CESAR: approx **5K*** events/spill
 - Trigger rate from TDAQ: approx **1.8K** events/spill
 - XCET44: pressure **0.308** bar, events **~300**, gas **CO2**
 - XCET48: pressure **1.005** bar, events **~5000**, gas **CO2**
 - Run started: **2026-08-14 23:47:18**
 - Run finished: **2026-08-14 23:58:23**
- **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **51.414**
- DESY table vertical position: **15.768**
- **Observation:** CAL13 IS CENTERED AROUND **1600** QDC COUNT.

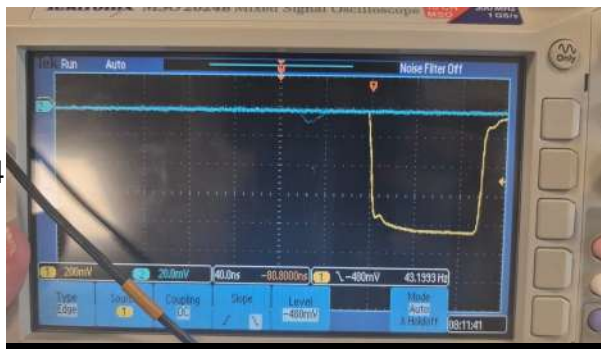
2026.08.13

As discussed we will first bring the evttrg and central 4 calorimeters to the zone. Central 4 calorimeters are 2 4 14 18. EVTTRG is at B01, CAL2 is at B02, cal4 is at b03 cal14 is at b04 and cal18 is at b05. to all the distances that they go into their modules, we have added 8 ns for everything and 13 ns for the cal18 because its unit is far away.

Okay, to here we are showing how much each of them is away from the trigger (EVTTRG is yellow, cal signal is blue).



CAL14



CAL18

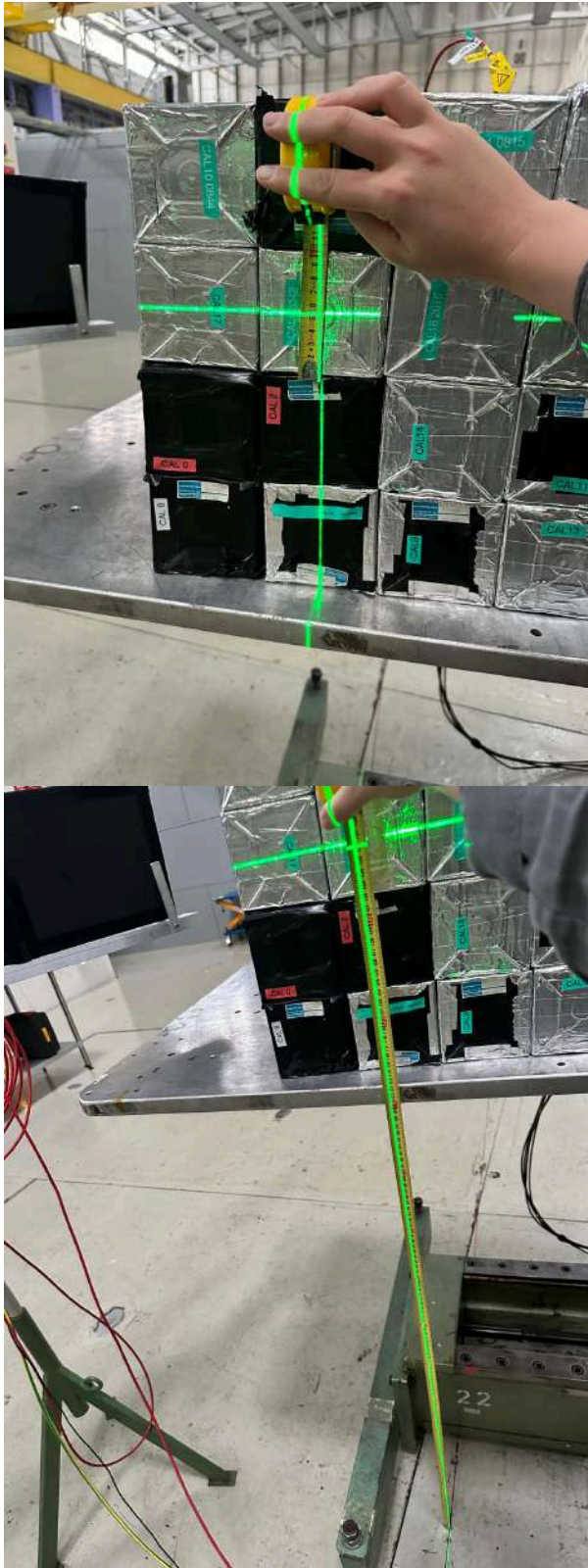


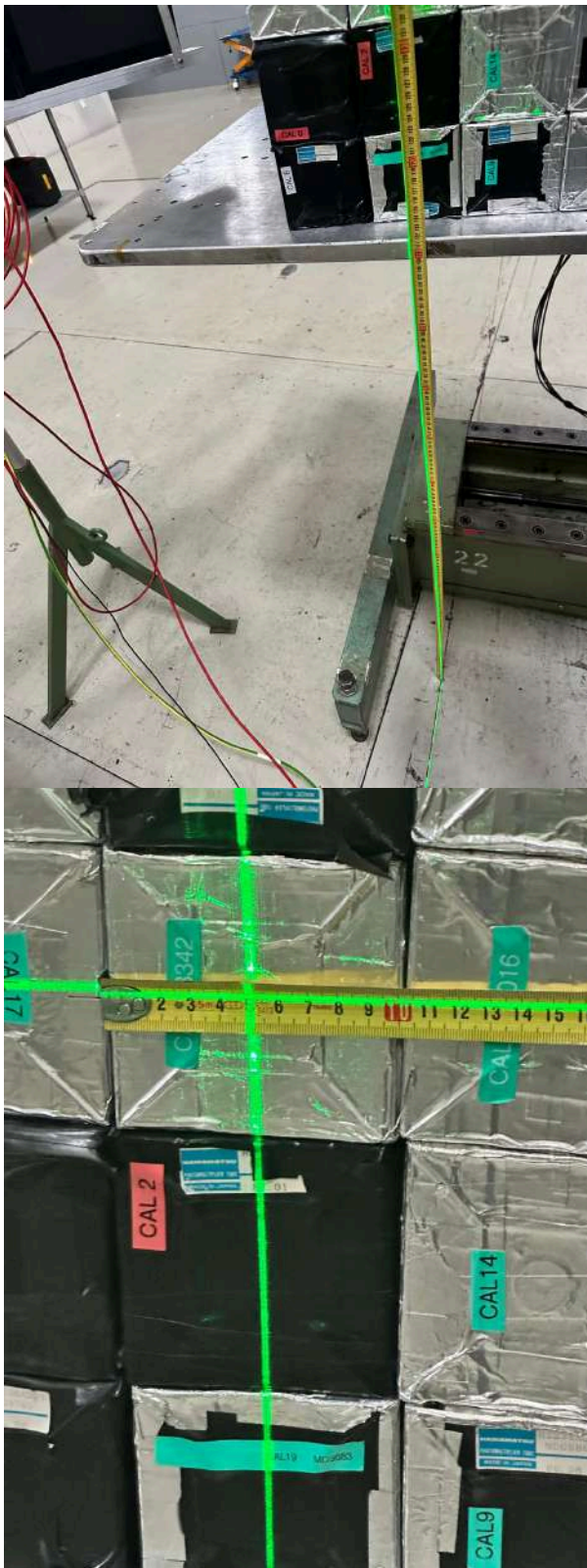
As it can be seen in here, some of them are 80ns away after the brown and additional 16 ns cables, some of them are more like 40 ns away. What we have decided on is that we are going to add 10 ns or 16 ns cables (we have 8 left of each) and apply 66 ns of delay unit's delay on top of this. but, as we are scared group of people, we will also make EVTTRG width to 120ns.

Here is how much ns is added to each of the calorimeters.

Cal	Delay (ns)
Cal0	10
Cal1	16
Cal2	16
Cal4	16
Cal5	16
Cal7	10
Cal8	16
Cal9	16
Cal10	10
Cal11	16
Cal12	10
Cal13	10
Cal14	16
Cal17	10
Cal18	10
Cal19	10

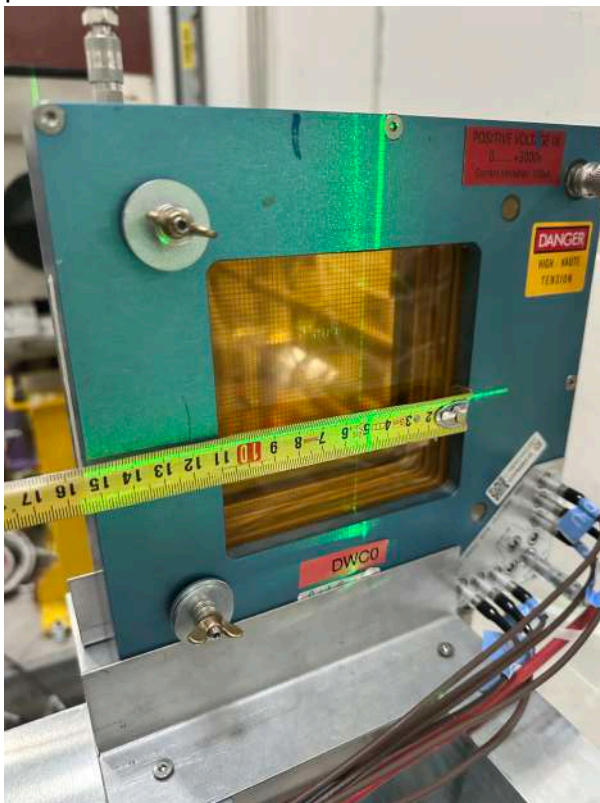
And then we have worked on the alignment of the laser lines to the Cal4. The DESY table values at this point is: H:31.472, V:-3.696.

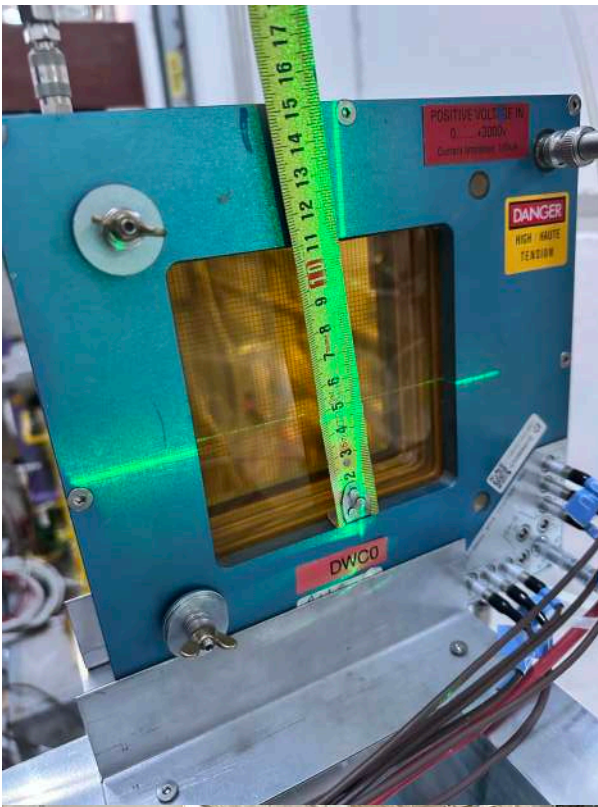






We have also tried to align the DWCs and the DWC1 is aligned well, DWC0 is not so much, pictures are attached.





and the XSCA table values at this points are V:3.96, H:0.80.

In principle, we are ready to turn on the detectors and start data taking. We leave the DWC calibration to tomorrow (we are tired, shocking we know), but we want to do the Cal4 calibration, and for this we will do with our previous HV values, because it seems the values that has been tested in the lab are too small.

[illegible]

- Run 1786657998 : -1 GeV BEAM, TRYING TO CALIBRATE CAL4

- **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **31.472**
- DESY table vertical position: **-3.696**
- **Observation:** XCET44 IS MISSING!!! CAL4 IS CENTERED AROUND 1100 QDC COUNT. It is interesting but on our scaler we can read out C0. We have increased the pressure on C1 to see how much of the beam is electron to 1 bar, it seems super high purity. we have recorded another file with all the same conditions but with 1 bar C0, beam file number 1786660465.
- Run **1786660965** : **-3 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **002**
 - beam momentum: **-3 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **74323**
 - **Initial look:**
 - Event rate from CESAR: approx **5556** events/spill
 - Trigger rate from TDAQ: approx **1800** events/spill
 - XCET44: pressure <value> bar, events <value>, gas <gas>
 - XCET48: pressure **1** bar, events **5159**, gas **CO2**
 - Run started: **2026-08-14 00:42:45**
 - Run finished: **2026-08-14 01:02:52**
 - **Calorimeters and DESY table:**
 - calorimeter gap: **NO GAP**
 - DESY table horizontal position: **31.472**
 - DESY table vertical position: **-3.696**
 - **Observation:** XCET44 IS MISSING!!! CAL4 IS CENTERED AROUND 3300 QDC COUNT.
- Run **1786662364** : **-2 GeV BEAM, TRYING TO CALIBRATE CAL4**
 - beamfile: **001**
 - beam momentum: **-2 GeV**
 - Trigger Condition: **S0S1**
 - Number of events: **1.86m**

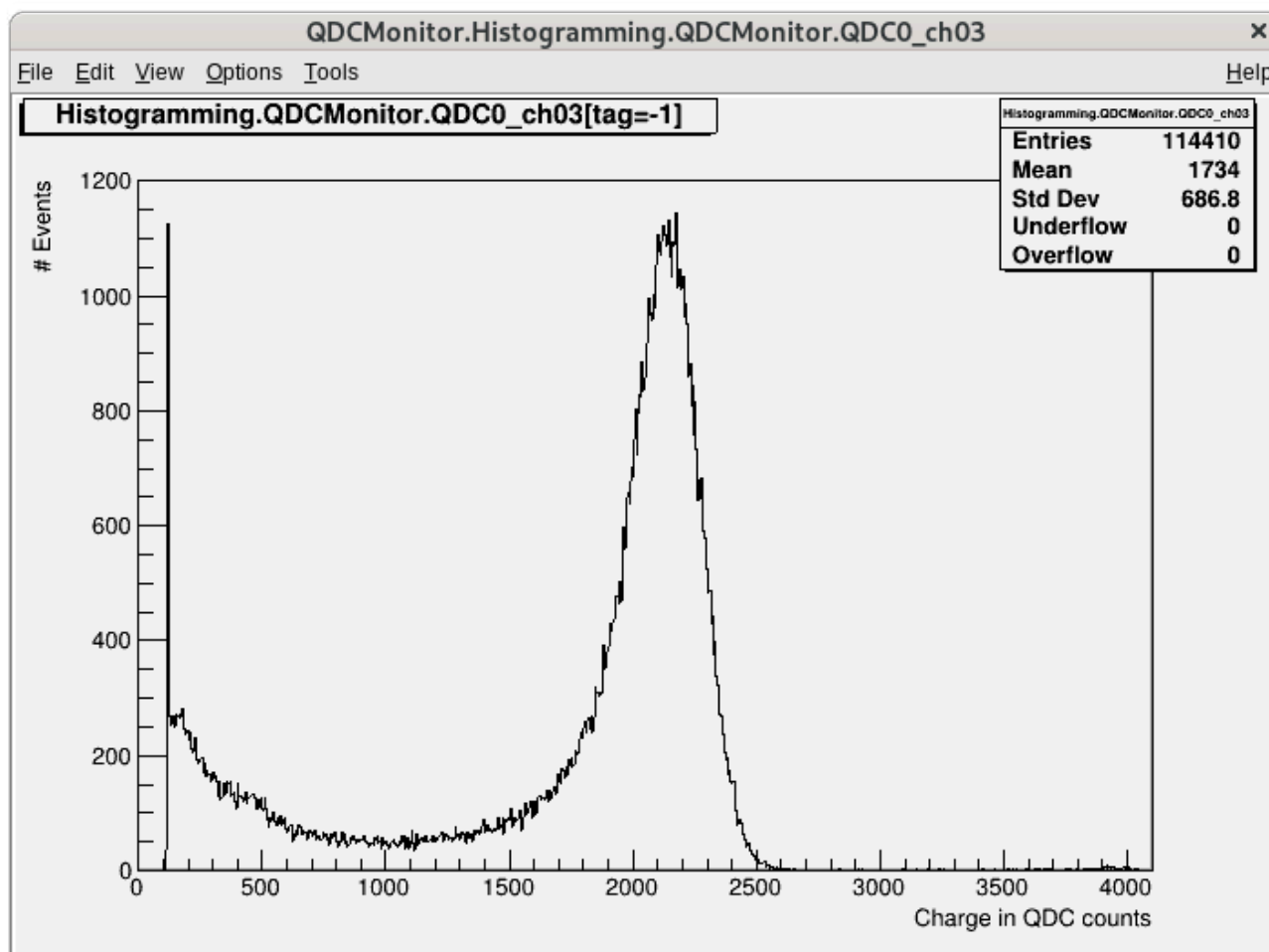
○ **Initial look:**

- Event rate from CESAR: approx **** events/spill
- Trigger rate from TDAQ: approx **1700** events/spill
- XCET44: pressure ?? bar, events ?, gas **CO2**
- XCET48: pressure **1** bar, events **5159**, gas **CO2**
- Run started: **2026-08-14 01:06:04**
- Run finished: **2026-08-14 09:29:13**

○ **Calorimeters and DESY table:**

- calorimeter gap: **NO GAP**
- DESY table horizontal position: **31.472**
- DESY table vertical position: **-3.696**

- **Observation:** XCET44 IS MISSING!!! CAL4 IS CENTERED AROUND 1734 QDC COUNT. Here is the SS below to see QDC counts from CAL4:



2026.08.12

18:38, we have finished installing all the detectors, cabling is not completely finished, safety check is done, and our dear beamline physicist Dipanwita will soon create our beam files, and we will start turning on our detectors on by one.

18:58 she has started.

19:25 Dipanwita is finished.

Afterwards we have done so much cabling, we did the calorimeters, DWCs, NIMs VMEs, and our NIM wiring cable and the VME layout is finalized, it can be seen from above.

AND THEN WE'VE HAD OUR FIRST TRIGGER WITH S0S1 with e-beam!!! We didn't save the file because there is no HV. We cannot see the IP of HV source.

 To get information of Main Frame IP ping the name of the device, for example ping POOL04310028.

We have made planning for tomorrow: firstly we will join our students' meetings, and then we will come back to the zone, first Berare will bring the signals to the zone (four central calorimeters) and the event trigger, and we will measure how far apart, and Seyma will do her presentation. And then Berare will do her presentation, Seyma will find out the central points of the calorimeters.

2026.08.11

Madeleine has measured the delay of each unit for us

Unit name:	Delay:
Discriminator 621 CL	6 ns
Coincidence 465	21 - 22 ns
Amplifier 612 A	14 ns
Fan-in/ fan-out	13-14 ns
Logic unit (dis)	45 ns
Logic Unit (coin)	80 ns
Logic Unit (fan)	55 ns

We have setup CAENGECO to bl4sdaq7, it automatically starts working when you should say caengeco2020 (it is part of the path). We are making the connection between bl4sdaq7, and sbc-isotdaq-5.

We are setting up the system on the opt mode.